Taking **Notes** & Giving **Updates** & Asking **Questions**

TIMESTEP apprenticeship 25-26

Research Notes

Why spend time discussing **note taking?**

Don’t you take them for classes already?

- **Research** notes serve a different function than **class** notes
- **Class** notes have a set narrative
- **Research** notes are an open and ongoing investigation

Things I had to learn the hard way

- **Code documentation.**
  Great! I wrote code for this problem 6 months ago! Oh but how does it work again...
  (spends half a day relearning own code)

- **Bug documentation.**
  Oh I've seen this bug before! But wait, how did I fix it...
  (spends an hour on stack overflow, again)

- **Clear calibration/reduction documentation.**
  I've been using these cleaned images for months, but how on earth did I process this data...

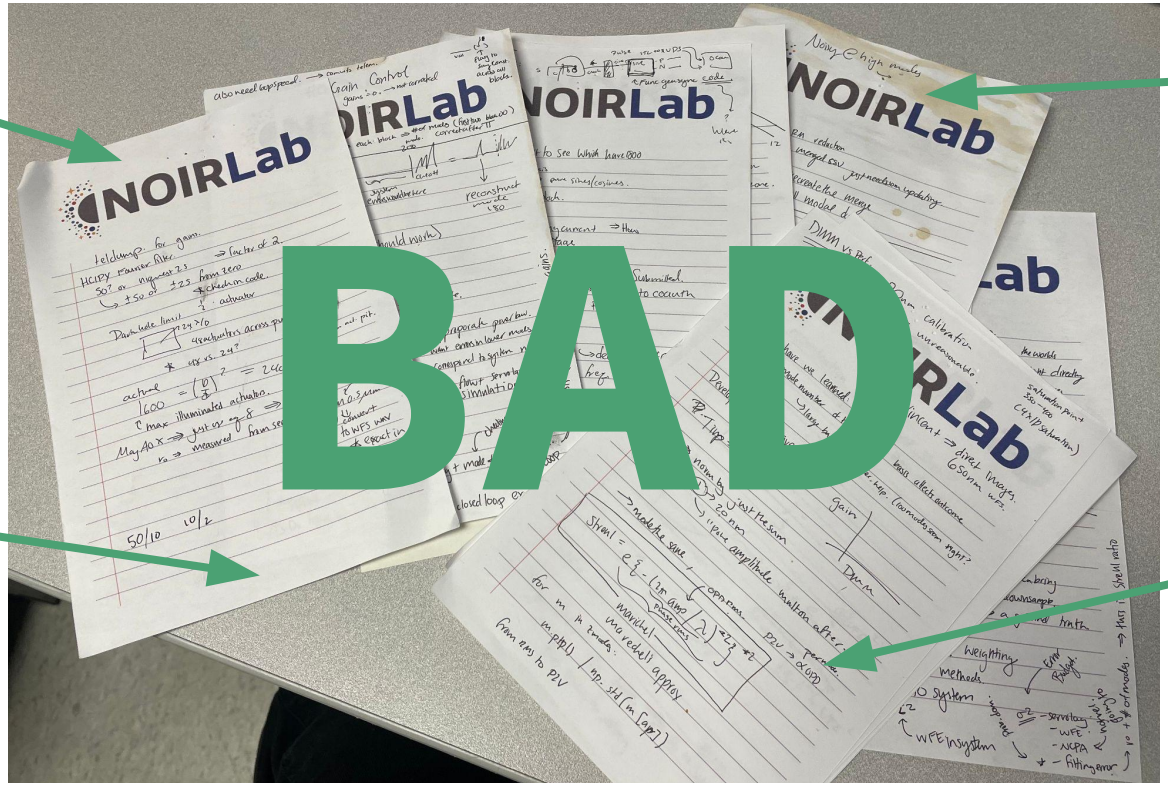
- **Clear plot labels/documentation.**
  Supervisor asks "What are the units on the wavefront aberration in this plot?". You made the plot last year. It has no caption.

What are they for?

- Keeping track of **what you did**
  - Papers read
  - Plots and their descriptions
  - Steps taken
  - Debugging solution
- Keeping track of **what you need to do**
  - Next steps
  - Papers to read
  - A bookmark for when you next get to research
- Keeping track of **Questions**
  - The foundation of research

Notes Examples

Eden - random slips of paper



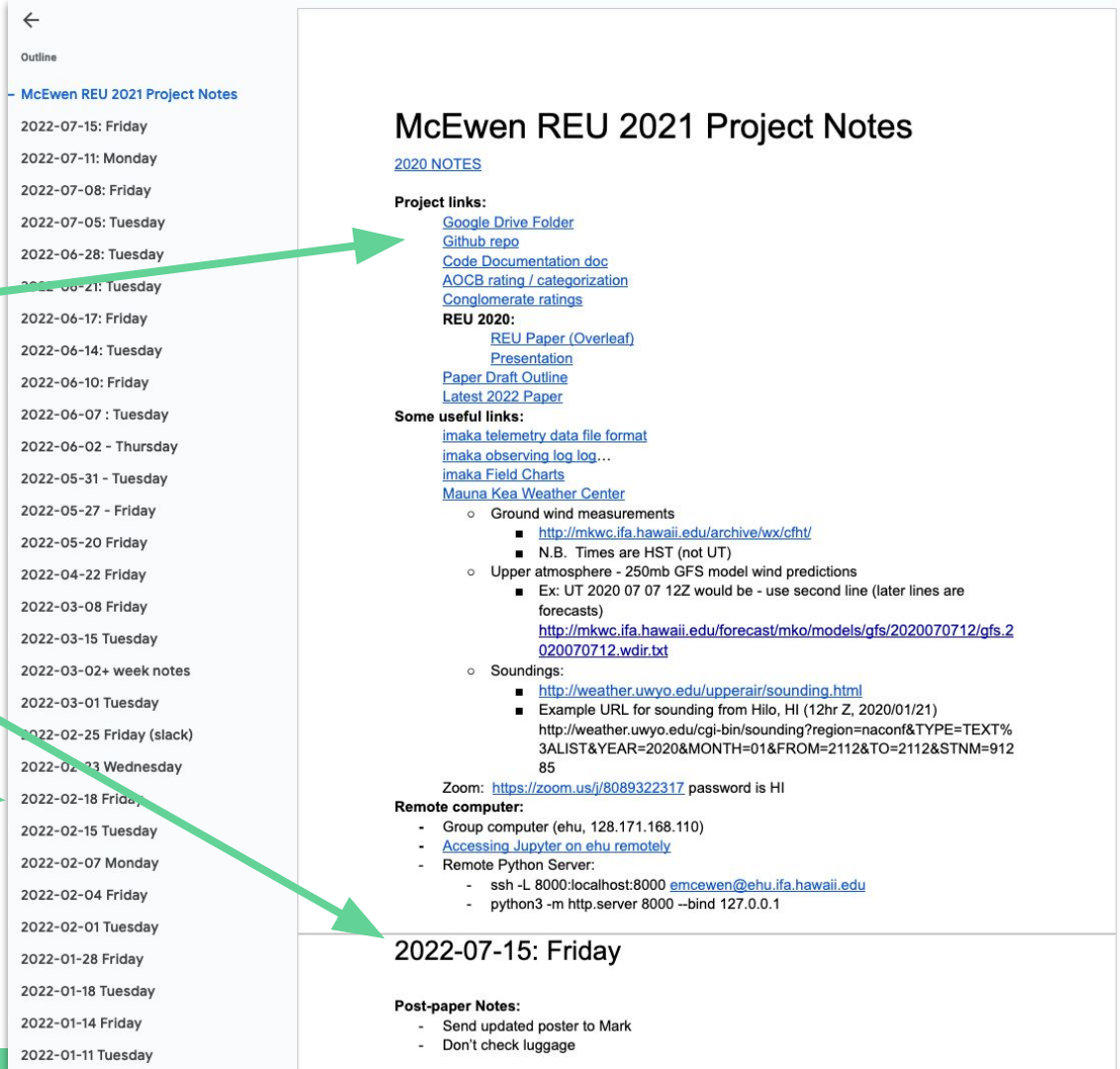
Digital note-taking

Eden - Google Docs

Links to frequently used sites / docs in header

Headers for date, most recent at top

Table of Contents autopopulates from header style entry



Eden - Google Docs

Plot, my update for the week

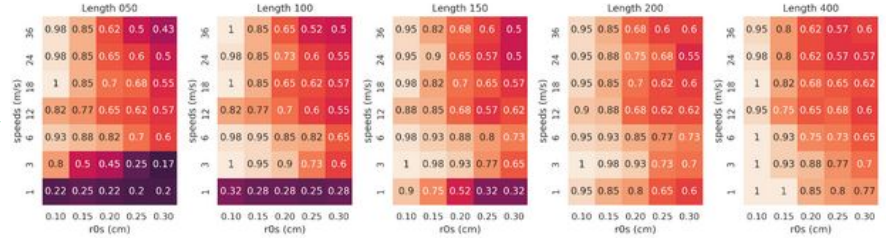
description, what I think this result means

Next steps, my guess at what I think I should try next

Question, what to ask my mentor at our next meeting

context, why I want to ask

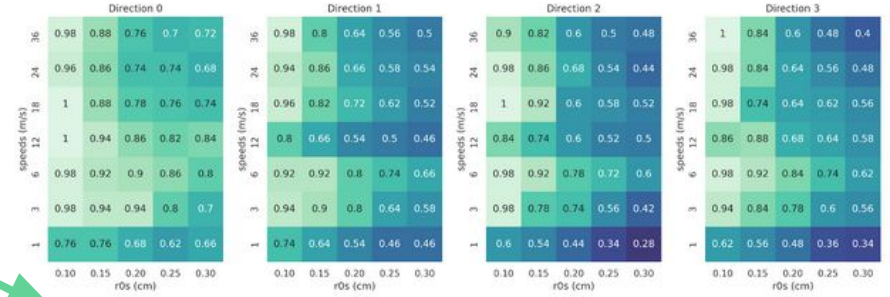
Plot: Hit Map update:



Conclusions:

- This might be too much information on a plot
- Conclusion: need a long enough window length for shorter layers
  - For this conclusion, might just need the last row.
- Conclusion: degradation of estimations with r0 (weaker)
  - Inject even weaker layers? 0.40 or more? What's the bounty edge of strength?
  - Additionally run through the next three increments of five

Directions:



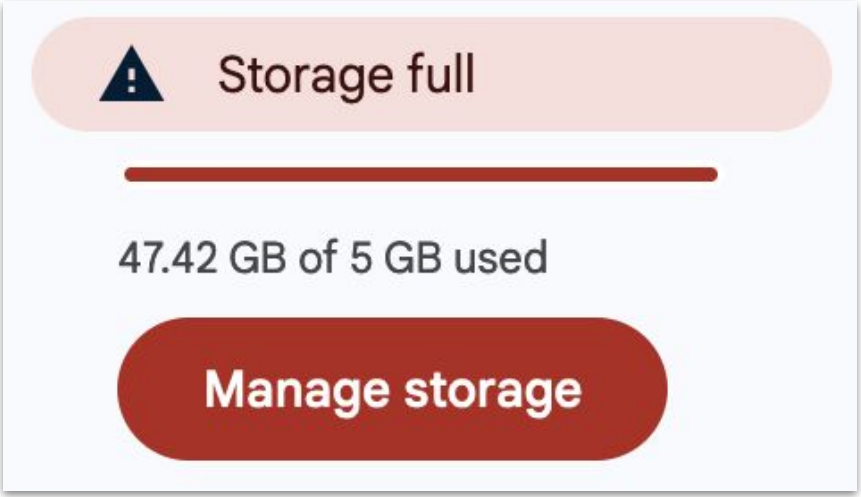
- How does a second layer contaminate these results?
- Base file giving us same speed and dir as the supposed "injected" layer
  - Can both "skew" or "override"
- Run: base files through analysis, see where they overlap in speed and dir

Eden - Google Docs

Why don’t I still use them?



… research notes stopped loading



Google pulled the plug on infinite storage

Meredith - python jupyter notebook

Timestamped reminders of changes, follows code progress

```
import sys
sys.path.append('~/Documents/')
import photometry_functions
```

```
datadir = os.getcwd()+'/J1120 Environment/data/'
```

```
# 2024-07-16: straight from quasar project
def get_wcs(directory):
    hdu = fits.open(directory)
    wcsi = wcs.WCS(hdu[1].header)
    hdu.close()

    return wcsi
```

```
# 2024-08-16: plotted apertures look wonky;
            # compile narrow-band and reference band (better SNR) source positions for matching
# 2024-08-19: results not great, switched reference band from F200W to F410M:
            # emphasize similar morphology to narrowband (+ SNR) over resolution
# 2024-08-19: switching back to using narrow-band for reference, F410M doesn't produce better apertures either

nb_dataref = datadir+'jw02774-o001_t002_nircam_f405n-f444w_i2d.fits'
nb_segmref = datadir+'jw02774-o001_t002_nircam_f405n-f444w_segm.fits'
mask = np.loadtxt('J1120 Environment/segm_mask.txt', dtype='bool')
nb_photref = photometry_functions.segment_phot(nb_dataref, nb_segmref, mask=None, plot_deblend=False) #maybe need to make mask

# wb_dataref = datadir+'jw02774-o001_t002_nircam_clear-f410m_i2d.fits'
# wb_segmref = datadir+'jw02774-o001_t002_nircam_clear-f410m_segm.fits'
# wb_photref = photometry_functions.segment_phot(wb_dataref, wb_segmref, mask=None, plot_deblend=False)
```

Meredith - python jupyter notebook

Markdown feature for notes, can do equations, adds to ToC on the side.

07-25-2024: trying to fix HST-JWST astrometry offset via aligning HST to JWST images: g

07-25-2024: JWST catalog looks good but resulting HST mosaic looks absolutely awful. align to Gaia/SDSS?

07-29-2024: not enough Gaia or SDSS sources to do robust matching. abandoning idea, commenting out

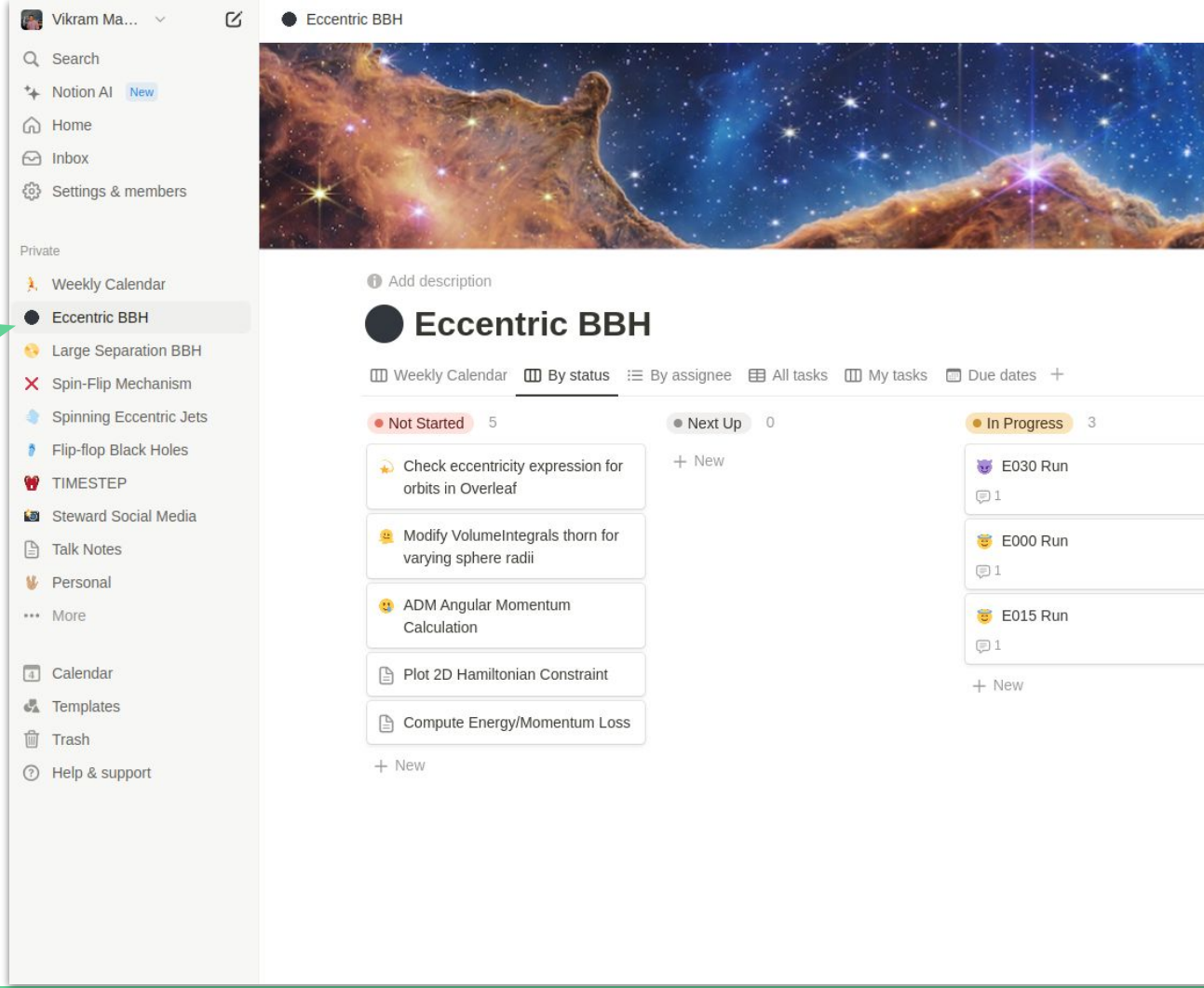
```
# # use f200w: high SNR and good resolution
# segment_200w = fits.getdata(os.getcwd()+'/J1120 Environment/data/jw02774-o001_t002_nircam_clear-f2
# data_shape = segment_200w.shape

# # masking locations duplicated from F405N source mask, should refine eventually b/c different pixe
# mask = np.zeros(data_shape, dtype='int')
# mask[2015:2213, 8015:8263] = 1
# mask[1008:1086, 5748:5833] = 1
# mask[2532:2613, 8086:8182] = 1
# mask[1874:2016, 1557:1745] = 1
# mask[1113:1200, 4204:4256] = 1
# mask[4285:4338, 2513:2572] = 1
# mask[0:4578, 0:100] = 1
# mask[0:4578, 4220:4360] = 1
# mask[0:4578, 5600:5747] = 1
# mask[0:4578, -100:-1] = 1
# mask[0:100, 0:10077] = 1
# mask[-100:-1, 0:10077] = 1

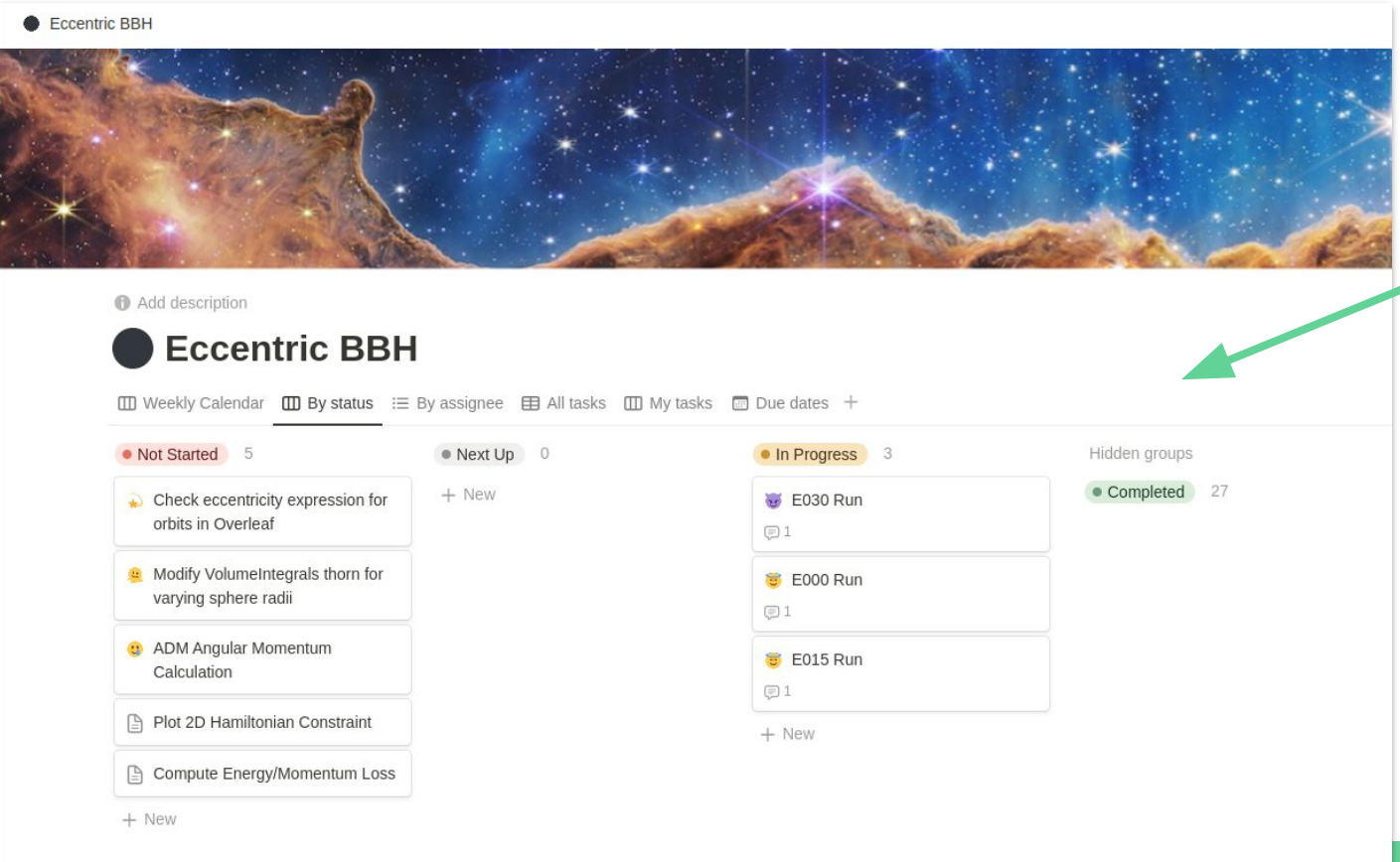
# np.savetxt('J1120 Environment/segm_mask_200w.txt', mask, fmt='%.1d')
```

Vikram - Notion

Separate projects
(and life!) on the side



Vikram - Notion



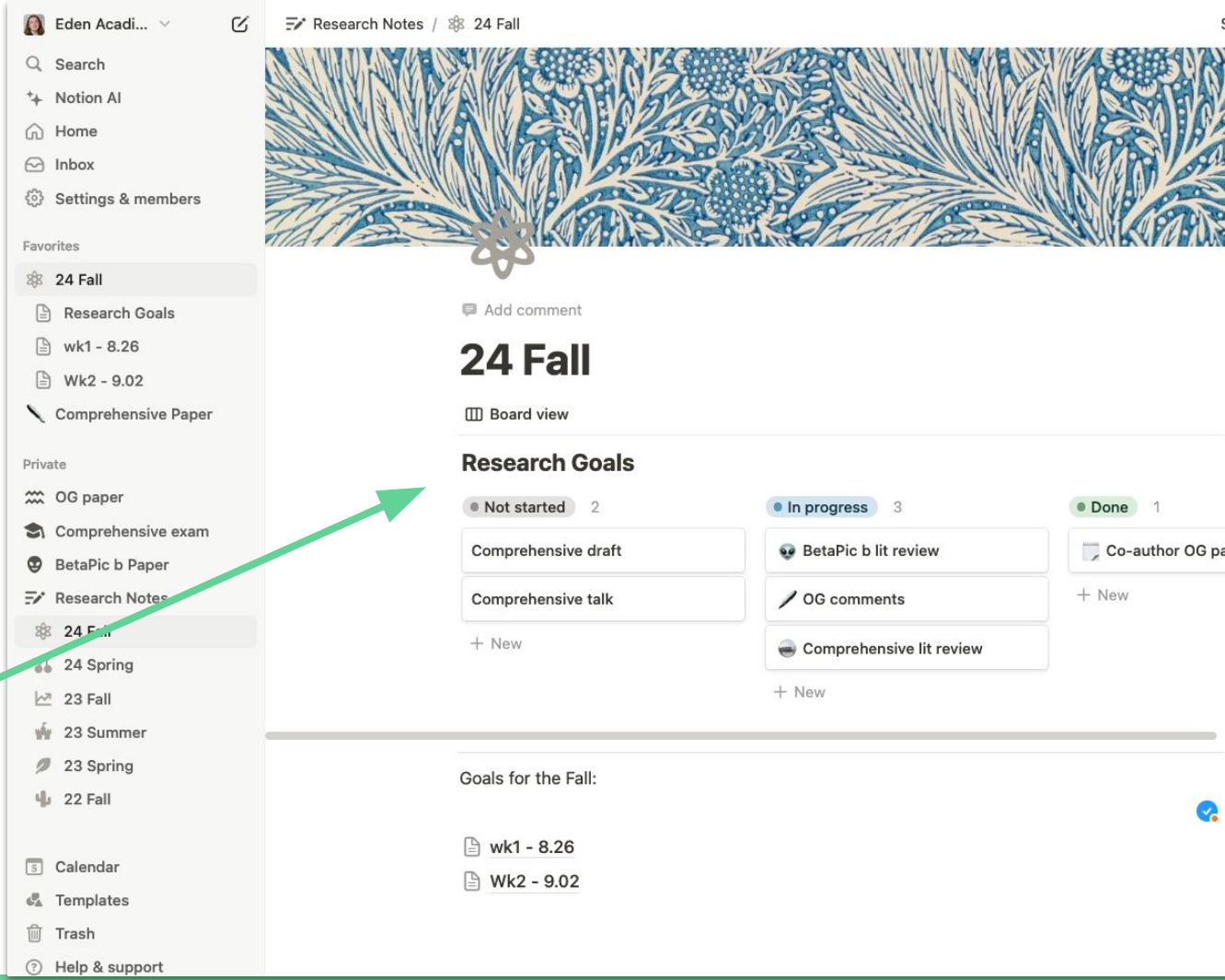
Kanban board
Integration, allows for tracking priorities and accomplishment

Eden - notion

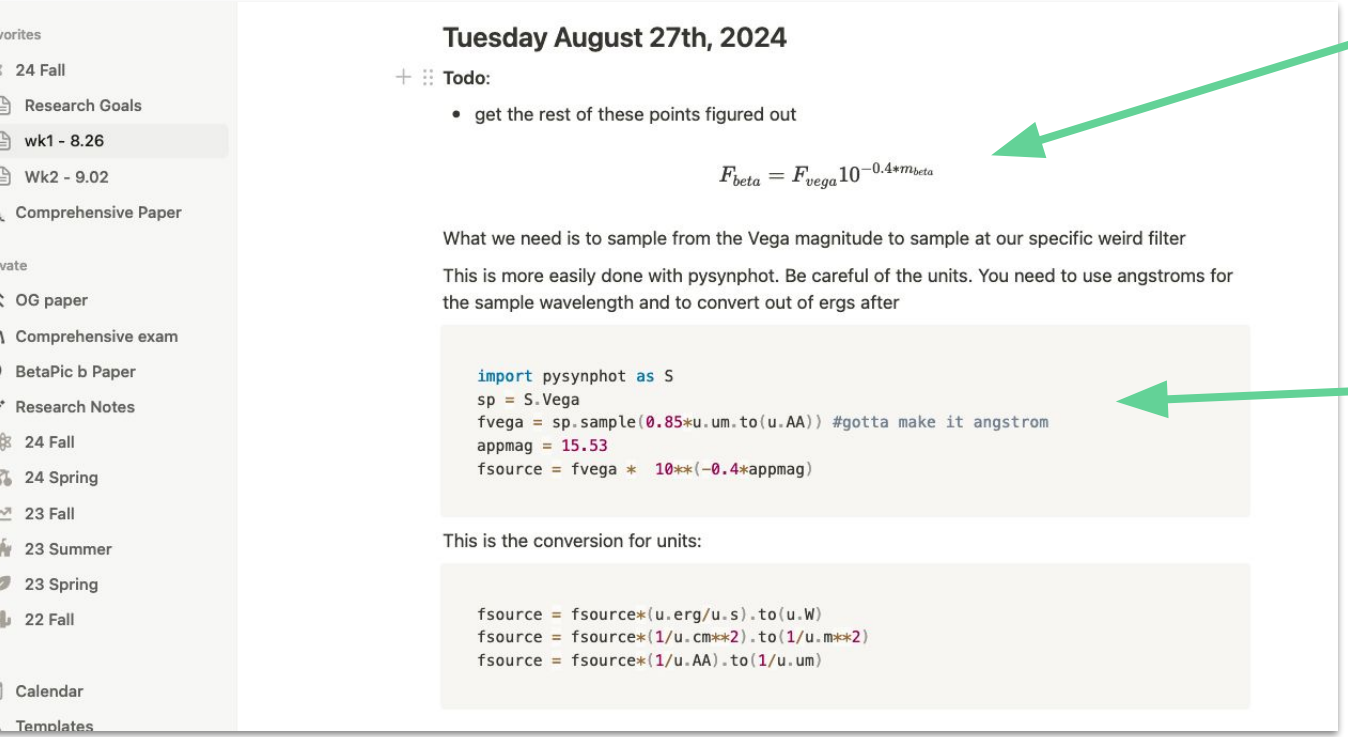
Weekly Pages,
for day to day debugging

Notes by semester,
for organization

Large Goals
highlighted on semester landing page



Eden - Notion



Latex Formulas

Copy and paste for your paper / official writeup

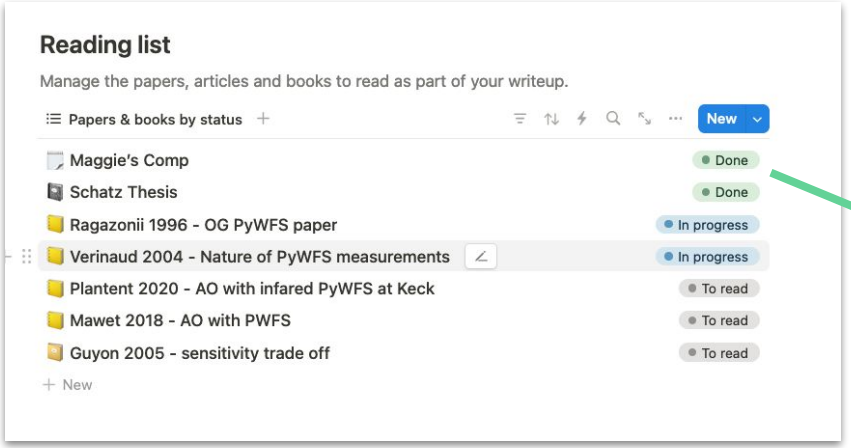
Code snippets

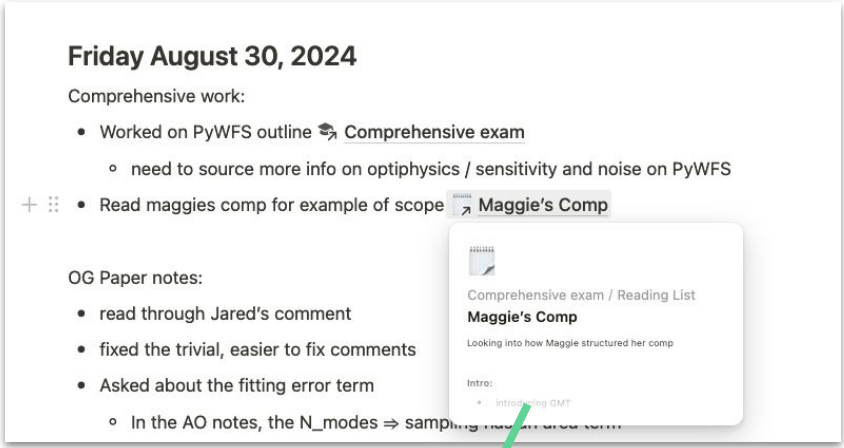
Well formulated for a visual aid

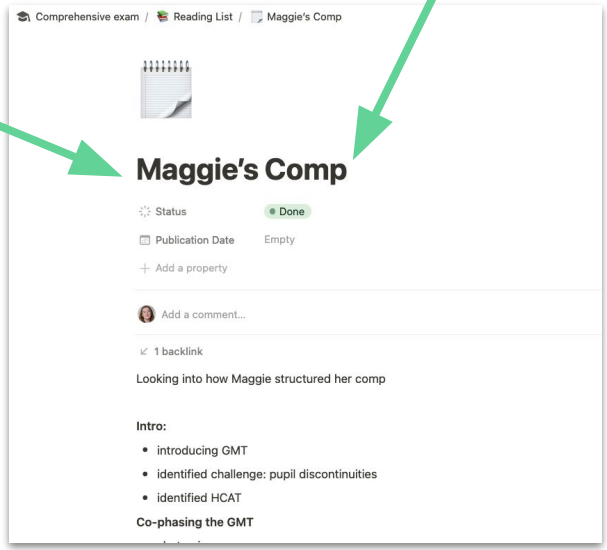
Helps when including a debugging stage

Eden - Notion

Link between pages, reference easily





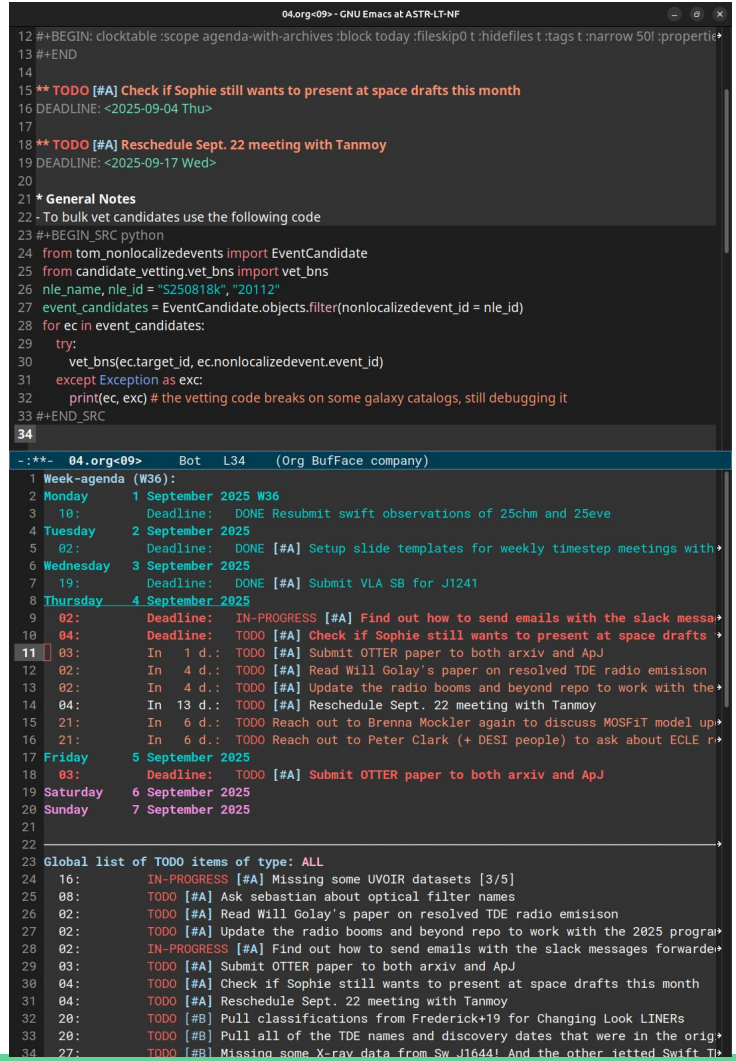


Noah - Emacs + org-mode

Advanced (Over-complicated?)

- One terminal command sets up a new daily research note file
- To Do items are automatically gathered into a single file for me
- *A lot* of overhead/setup/learning

Ask me (or Vikram!) if you're interested in more details



Physical note-taking

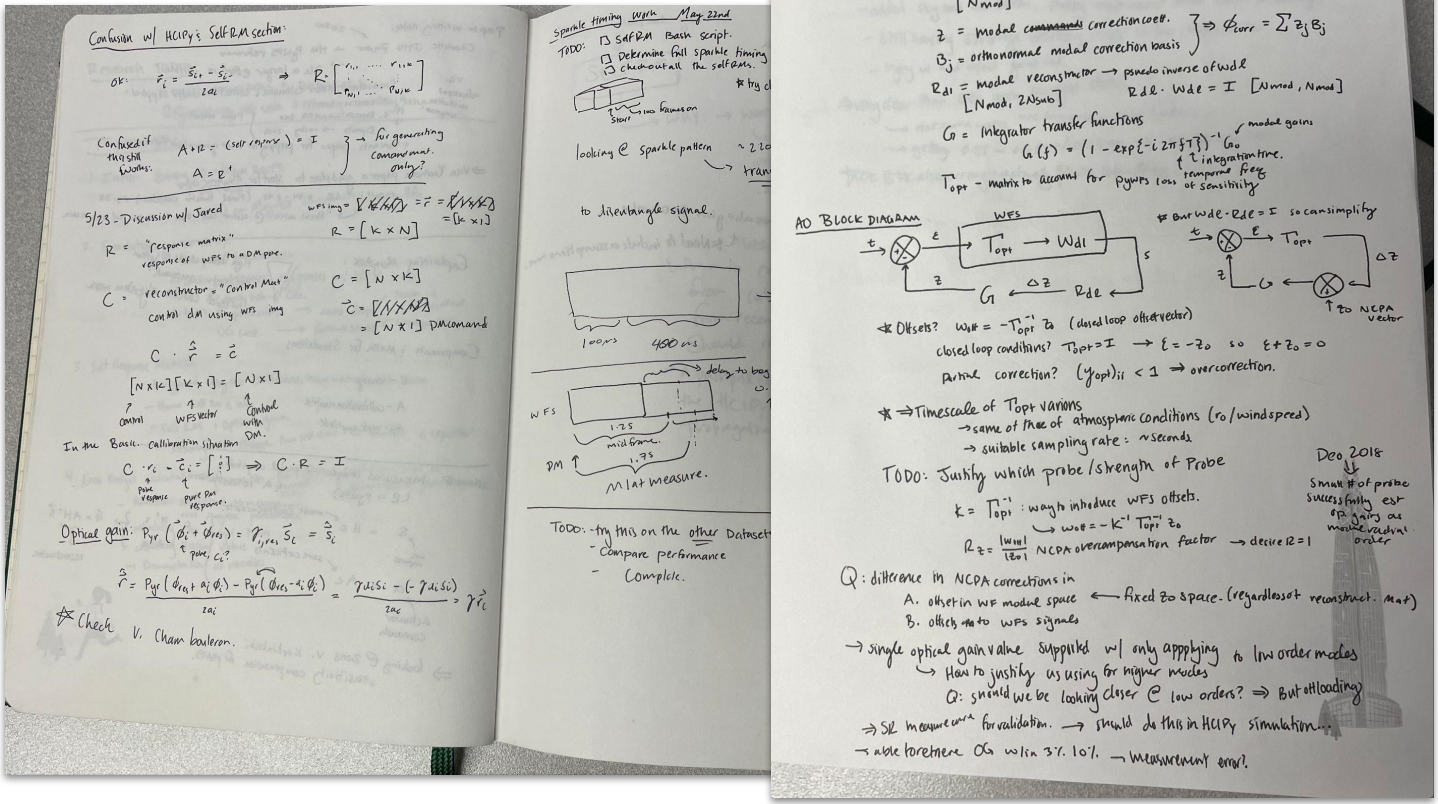
Personal notebooks

Meetings

Brainstorming

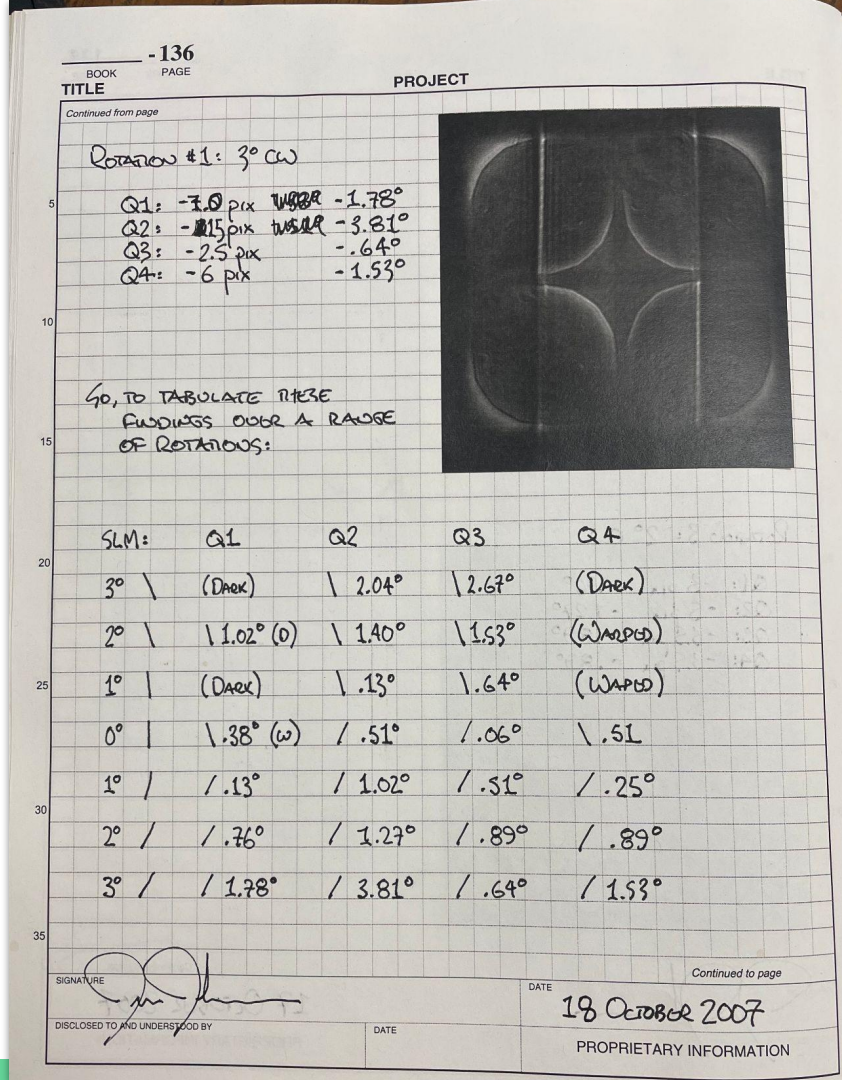
Sketching

math derivations



Lab notebooks

- Official Documentation
  - Not just notes for you
  - Should include name and date
- Taken while doing experiments
  - Treat your research better than scribbles on a napkin
- Information required to reproduce results
  - Historical account of the research
  - can’t just rerun like code



-136
BOOK PAGE
TITLE PROJECT
Continued from page

Rotation #1: 3° CW

Q1: -7.0 pix twist -1.78°
Q2: -6.5 pix twist -3.81°
Q3: -2.5 pix -.64°
Q4: -6 pix -1.53°

So, to tabulate these findings over a range of rotations:

SLM:	Q1	Q2	Q3	Q4
3° \	(Dark)	\ 2.04°	\ 2.67°	(Dark)
2° \	\ 1.02° (0)	\ 1.40°	\ 1.53°	(Warped)
1° |	(Dark)	\ .13°	\ .64°	(Warped)
0° |	\ .38° (0)	/ .51°	/ .06°	\ .51
1° /	/ .13°	/ 1.02°	/ .51°	/ .25°
2° /	/ .76°	/ 1.27°	/ .89°	/ .89°
3° /	/ 1.78°	/ 3.81°	/ .64°	/ 1.53°

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DISCLOSED TO AND UNDERSTOOD BY
DATE
18 October 2007
DATE
PROPRIETARY INFORMATION

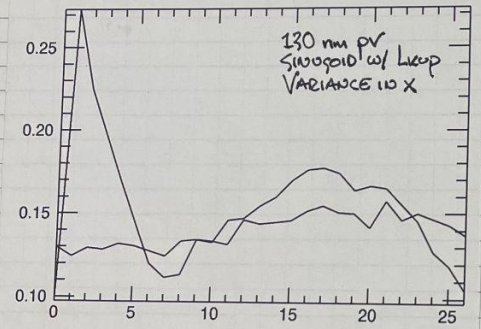
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30 July 2007

- Office Day. Reading & Paperwork

31 July 2007

- Going to register, align and focus, and do one last data collection before digging into hardware.
- I'm going to do a run in the middle of the range: 130 nm peak-to-valley. (130)
- This run is using lookup table, and produces this MTF:



- So this by-now predictible behavior - crashing out at ~ 1/4 Nyquist, no flat plateaus - is unfortunately replicating.

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- From the stored data, I'm interested in the intensity information from the pupils. The following is pupil analysis per iteration for the above run.

Iter	Min: 1	2	3	4	Max: 1	2	3	4	RMS: 1	RMS 2	3	4
1	5	8	6	7	52	62	44	38	23.4	32.8	17.8	19.9
	1	10	1	7	50	86	36	65	14.7	46.4	10.2	31.4
	2	3	2	2	85	71	75	59	31.3	32.3	21.5	22.0
	2	6	2	1	70	73	52	67	27.7	34.3	19.3	24.3
5	2	6	2	1	76	75	58	67	28.3	36.0	18.4	25.5
	2	4	2	2	85	81	66	68	31.1	33.6	19.5	24.9
	2	3	2	2	76	80	56	70	28.3	32.8	19.0	25.5
	1	5	82	2	73	71	53	68	28.0	25.6	19.0	26.8
.	1	4	3	2	79	74	62	67	30.0	33.9	18.7	25.9
10	0	5	3	2	74	75	57	70	29.7	33.7	17.3	26.5
	2	4	3	1	74	74	50	69	30.8	32.4	17.4	26.0
	1	5	3	1	73	69	57	61	29.9	32.8	17.7	23.3
	1	6	83	1	77	68	53	66	31.3	32.6	16.8	25.3
	0	5	4	1	73	66	46	58	31.1	32.6	16.8	26.2
15	2	6	4	1	65	68	53	62	28.5	32.3	16.6	23.6
	2	5	4	1	72	67	53	64	29.1	32.1	16.7	26.0
	1	7	4	2	67	66	47	62	29.8	33.3	16.0	26.0
	0	6	4	2	69	63	53	59	30.0	32.4	16.5	25.2
	1	7	84	2	78	65	51	60	30.7	32.5	16.3	25.6
20	2	7	85	1	76	64	48	59	30.0	31.3	17.1	25.2
	2	7	4	1	65	61	51	59	28.5	31.0	17.2	26.2
	1	7	84	1	71	64	47	61	30.3	32.9	15.4	25.3
	1	5	5	1	72	63	48	60	29.3	31.3	17.1	35.4
	2	7	5	1	74	62	51	57	30.2	32.1	16.8	24.3
25	2	8	5	2	73	61	48	63	30.2	32.0	16.6	23.6
	1	8	4	1	69	59	47	57	29.1	32.8	16.3	24.3
27	4	7	5	3	62	58	46	49	25.9	31.9	16.6	20.6

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Notebook images courtesy of Jess Johnson

Observing Log: a digital lab notebook

Similar idea - documenting trials, times, and anything relevant for those using data

25-26 03 2024 - The MagAO-X Observing Log

Observer	Catalog Name	Observation Name	UT start	UT end	seeing	cam sci1	cam sci2	cam wfs	cam lowf	cam tip	cam acq	Loop Freq	Modes closed	Comments			cam1 exp	cam1 EMgain
as logged	(ie HD799)	(ie speckle_calib_200Hz)			DIMM	Mark with x or filter value						(hz)		Note if loop is stopped, abnormal AO, LOWFS/Sp	Lowfs	sprkl		
Sebastiaan	betPic	fdpr_beta_pictoris_run_02	23:56:28	23:57:57	0.6	CH4	r					2000	1564	fdpr (0.9 - 0.95 phase-only, 0.95-amplitude only)				
		fdpr_beta_pictoris_run_02_unsats	23:58:30	23:58:00	0.65	CH4	r											
		fdpr_beta_pictoris_run_02_sats	23:59:00	00:00:00										NOTE: knifeedge left in				
														SLOW TURB NIGHT				
Sebastiaan	betPic	EFC_beta_pictoris_run_01	00:08:00	00:08:40	0.6	z	r	x	x	x		2000/2.	1564		1.8kHz, TTF			
		EFC_beta_pictoris_run_02	00:08:00	00:18:00	0.64	z	r	x	x	x		2000/2.	1564	"I'm probably saving all the DM channels" - 00:11	1.8kHz, TTF			
			00:18:00		0.45									iEFC On/Off Demo				
		EFC_beta_pictoris_run_03	00:19:00	0:26:50	0.45	z	r	x	x	x		2000/2.	1564		1.8kHz, TTF			
			00:26:00											Added in sparkles - 45deg, amp 0.02 20ld				
		EFC_beta_pictoris_run_04	00:27:00	00:39:00	0.45	z	r	x	x	x		2000/2.	1564	Long Run	1.8kHz	45, 0.02, 20		
			00:31:00											changing lowfs loop - pulling back?				
			00:38:00											LO closed - new reference				
		EFC_beta_pictoris_run_05	00:39:00	00:49:00	0.6	z	r	x	x	x		2000/2.	1564	now closed and happy		45, 0.02, 20		
		EFC_beta_pictoris_run_06	0:49:00	00:55:00	0.65	z	r	x	x	x		2000/2.	1564	Switching to high speed FPWS - 3kHz on LOWFS		45, 0.02, 20		
			0:52:00											Dark hole looks drifted - zeroed loloop - unclear performance				
			0:52:00											Lowfs turned back on				
		EFC_beta_pictoris_run_unsats_psf_cer	0:56:50	00:57:17	0.69	z	r	x	x	x		2000/2.	1564			45, 0.02, 20		
		EFC_beta_pictoris_run_sats_psf_centr	0:57:30	00:58:04	0.69	z	r	x	x	x		2000/2.	1564	NOTE: Sparkles are aligned with spiders :/ not great		45, 0.02, 20		
		EFC_beta_pictoris_run_07	1:00:00	01:13:00	0.62	z	r	x	x	x		2000/2.	1564		1.98 kH	45, 0.02, 20		
			1:13:00		0.40									pyWFS ran away, looked like a TT overdrive				
			1:13:00											Lost alignement, had to move camlens while CLloop was open				
		EFC_beta_pictoris_run_08	1:17:00	01:27:00	0.45	z	r	x	x	x		2000/2.	1564	NOTE: camsci 2 was closed this whole time (Eden tried to tell people but they ignored her)				
		EFC_beta_pictoris_run_08_unsats	1:28:00	01:29:00	0.52	z	r	x	x	x		2000/2.	1564					

Note-taking

Summary

Notes Philosophy

- Don’t just put successes, also **include failures**
  - Detail what mistakes you made to get there, how to get out of it
  - will often keep you from making the mistake again

- **Don’t assume you’ll remember** why you included something
  - Be kind to yourself, give yourself context when including plots / screenshot etc
  - Prepping a summary while you’re fully in it will help you when you explain to your mentor

- Make this a **resource** for yourself
  - Label your plots when you make them
  - Describe your plots, make them searchable (cmd-F)
  - Throw in the resources you used to make progress (links, websites, etc.)

How do you take notes?

What has worked?
What hasn't worked?
What do you want to try?

Discuss with your groups.

Research slides

Research slides?

What - a weekly slide update

Who - your research advisor 1:1, research group meetings, TIMESTEP grad meeting

Why - quickly summarize updates, communicate research process

What should a weekly research slide contain?

- Name and date
- Work accomplished that week
  - Sometimes this looks like a plot
  - Sometimes this is “reviewed literature”
  - Debugging counts
- Context for audience / future you
  - I can scroll back through my weekly updates to figure out what I was up to certain weeks
- Questions
  - Questions about work
  - Blockers - things blocking progress

This list should look like our list for notes

TIMESTEP
slide template

SLIDE TEMPLATE: Week of 01/01/2001

Updates on work done last week:

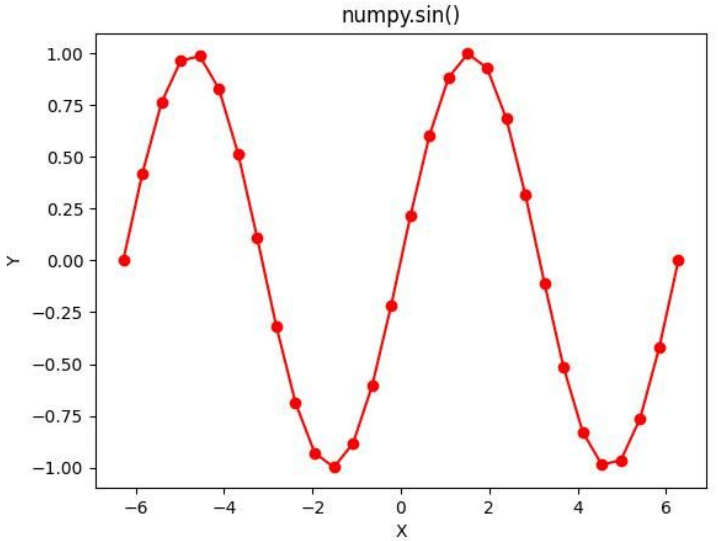
- Task with details
- Task with details

Current roadblocks/questions:

- roadblock with details
- question with details

Next week's steps/goals:

- Goal or next step
- Goal or next step



<Relevant figure with detailed caption. This can be a data plot, picture of lab work, code snippets, or components working with.>

Example slides

Eden

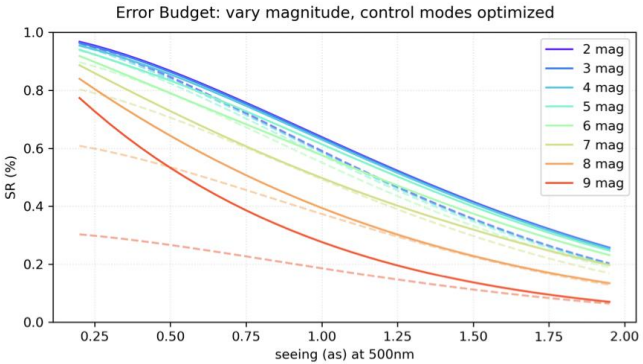
Sometimes it’ll be victorious

The week all my paper plots were finalized

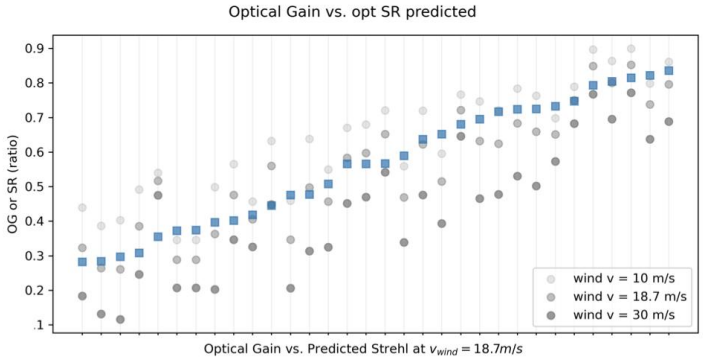
Eden McEwen
06.07.2024

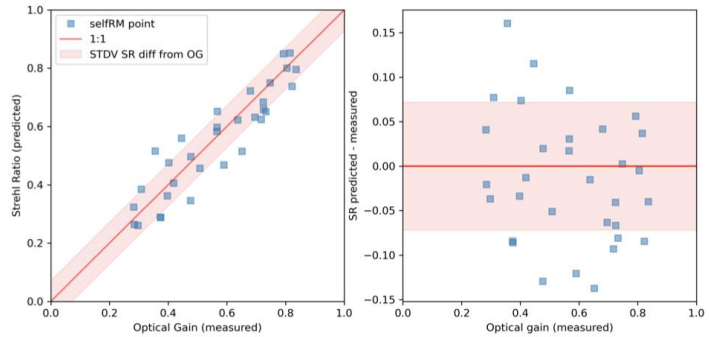
sparkles

- 2nd draft of JATIS paper done
- Updated the way the error budget handles modes controlled, things look better now
- Started cacao simulations on computer to get OG process developed
- Working on SPIE presentation: Practice monday?



Solid: number of modes we SHOULD have used
Dashed: Number of modes we used (1564)





Optical gain follows Strehl ratio 1:1

Sometimes it won’t be

(Spent all week trying to connect to a remote computer…)

I was feeling angsty about this

Could have been more descriptive of next steps

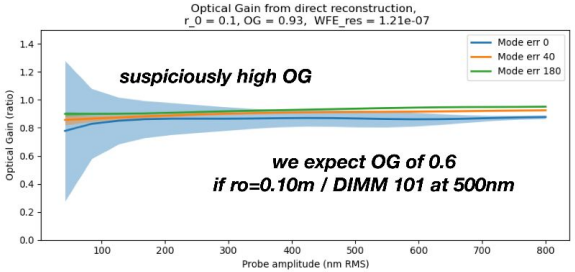


Typical grad level weekly update

Eden McEwen
09.20.2024

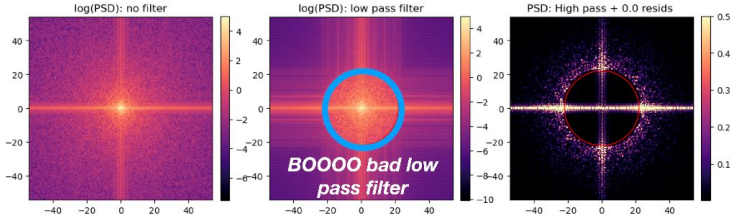
Optical Gain

- **Fighting the good fight with FFT**
- Have to adjust the way I'm creating turbulence for the paper
- having trouble with the filtering of lower modes that I hadn't with my previous method
- its def a function of hen the aperture is applied.
- a problem bc the r_o to residual WFE isn't lining up

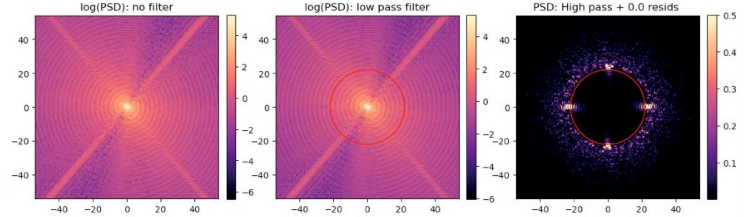


Other than that…

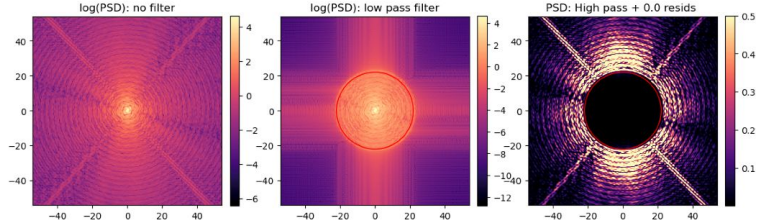
- Read a bunch of Comprehensives and PyWFS papers
- wrote a section of my comp
- thought wistfully about beta pic as I did more atmosphere simulations



How I'm trying to filter out low order modes now, the FT I'm using to show the PSDs has the cross artifact from the square discrete image (no aperture multiplication



If I multiply in the aperture right before the psd like (phase_filter*aperture), get residuals



the phase gets multiplied by the aperture before any filtering occurs. still see a cross in lowpass filter output.

Example slides

Vikram

Why and How does accretion happen?

Viscosity:

$$\nu = \alpha_{SS} H C_s \qquad C_s = \frac{H}{R} v_k$$

Radial Velocity:

$$v_r = -\frac{3}{2}\frac{\nu}{R} = -\frac{3}{2}\frac{\alpha_{SS} H C_s}{R}$$

$$\Rightarrow v_r = -\alpha_{eff}\left(\frac{H}{R}\right)^2 v_k$$

Since v_r is measurable, we can calculate α_{eff} → what α needs be given the mass accretion rate.

α is made up of two components within the disk:

$$\textit{Reynolds Stresses: } \alpha_R = \frac{\rho < u_r u_\varphi >}{< p_{tot} >}$$

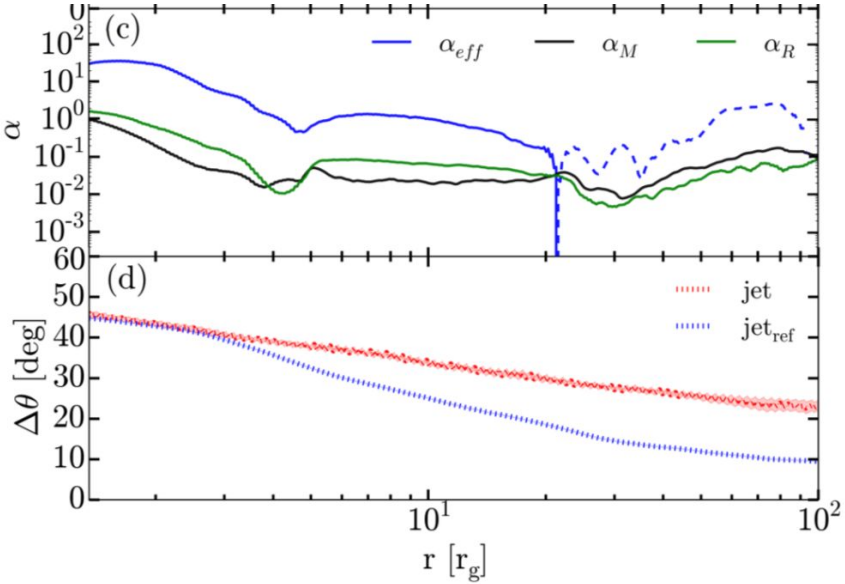
$$\textit{Magnetic Stresses: } \alpha_M = \frac{< B_r B_\varphi >}{< p_{tot} >}$$

~~*we expect:* $\alpha_{eff} = \alpha_R^x + \alpha_M^x$~~

$$\alpha_{eff} > \alpha_R + \alpha_M$$

The mass accretion rate is much greater than local Reynolds and Magnetic stresses in the disk can account for.

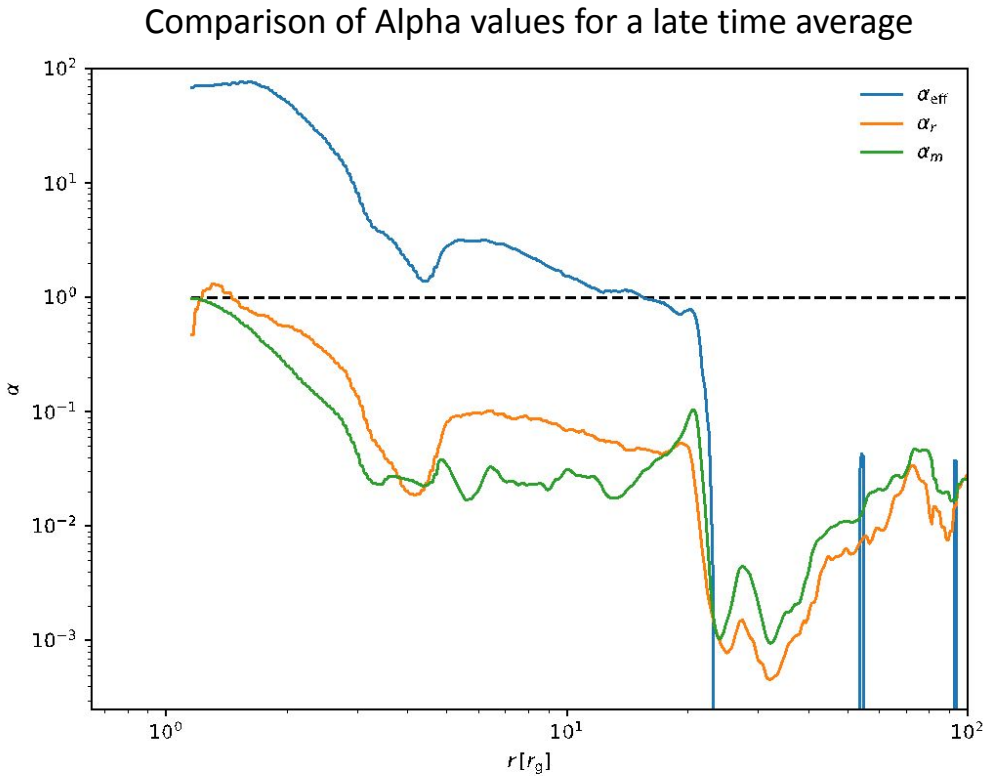
Why and How does accretion happen?



Only the top figure is relevant, look how α_{eff} is an order of magnitude larger than α_R or α_M [3]

[3] Liska, M., et al. "Bardeen–Petterson alignment, jets, and magnetic truncation in GRMHD simulations of tilted thin accretion discs." *Monthly Notices of the Royal Astronomical Society* 487.1 (2019): 550-561.

Plots


Comparison of Alpha values for a late time average
$\alpha_{\rm eff}$
α_r
α_m
α
$r[r_g]$
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^0
10^1
10^2

Calculating the Stress-Energy-Momentum Tensor T^{μ}_{ν}

$$\alpha^{\mu}_{\nu} = \frac{T^{\mu}_{\nu}}{p_{tot}}$$

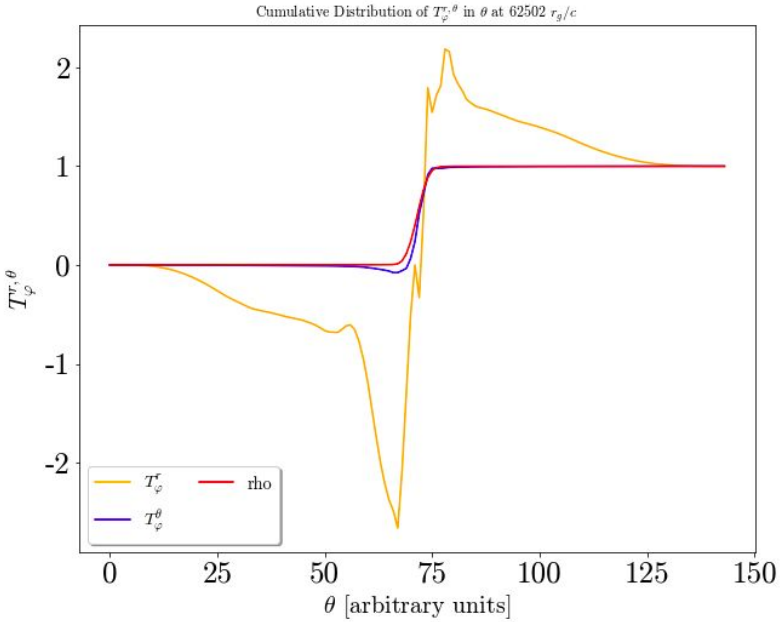
$$T^{\mu\nu}_{\mathrm{MHD}} = (\rho + u + p + b^2)u^{\mu}u^{\nu} + (p + \frac{1}{2}b^2)g^{\mu\nu} - b^{\mu}b^{\nu} \qquad (11)$$

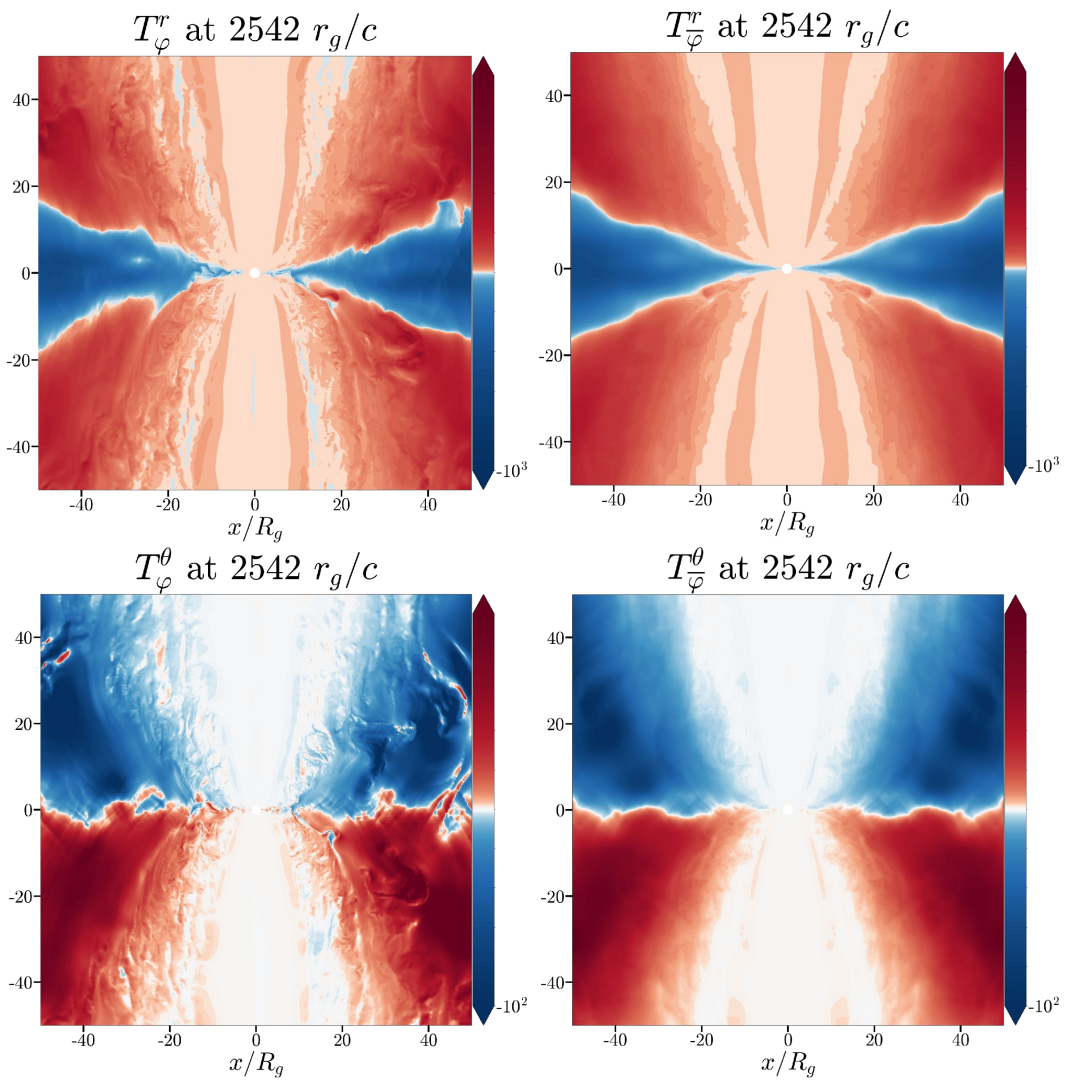
Note: we're using T^{μ}_{ν}

```
for i in range(4):
    for j in range(4):
        T_r[i][j] = gdet*((rho + ug + (gam - 1)*ug + bsq)*uu_[i]*ud_[j])
        T_m[i][j] = -gdet*(bu_[i]*bd_[j])
        T_g[i][j] = gdet*(((gam-1)*ug + 0.5*bsq)*kronecker[j][i])
        T_adv[i][j] = gdet*(( rho + ug + (gam - 1)*ug + bsq)*vd_avg[j]*vu_avg[i])
        T_turb[i][j] = gdet*(( rho + ug + (gam - 1)*ug + bsq)*(ud_[j]-vd_avg[j])*(uu_[i]-vu_avg[j]))
T = T_r + T_m + T_g
```

My Question: How should I visualize the data?

- Visual Density Plots
  - Time and phi averaged
- Visual Streamline plots
- Cumulative plots across radii/angles
- Something else?
- Movie – how it gets to steady state

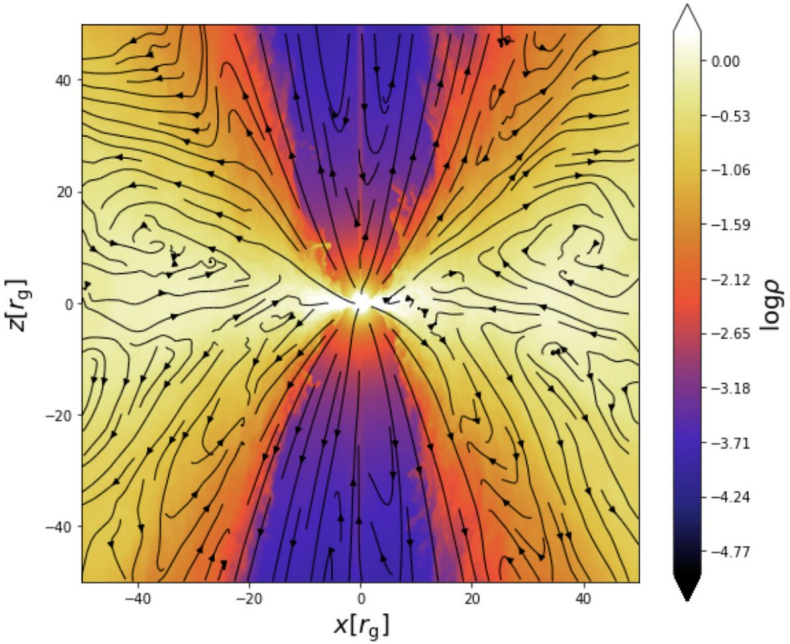


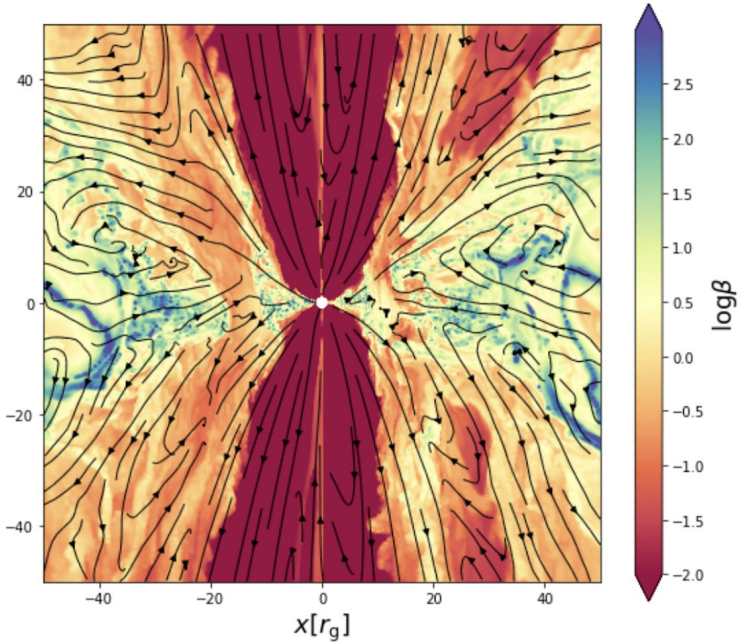

T^r_φ at 2542 r_g/c
$T^r_{\overline{\varphi}}$ at 2542 r_g/c
T^θ_φ at 2542 r_g/c
$T^\theta_{\overline{\varphi}}$ at 2542 r_g/c
40
20
0
-20
-40
-40
-20
0
20
40
x/R_g
-10^3
-10^2

8/27/21: Updates

- Streamlines
- Theta Profiles

Time averaged 7500-10000 rg





Todo:

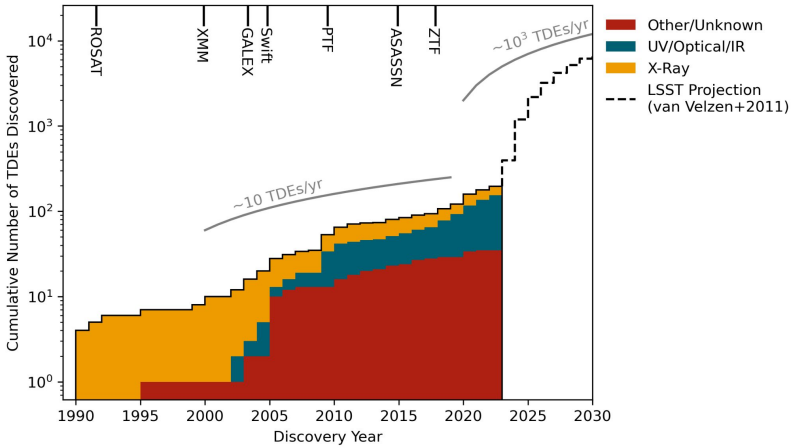
- Finalize theta profiles
- Finalize time averages
- Alpha effective plots – alpha effective density plot maybe???
- Plot for other simulations – TOR1024,
- Read literature and write paper

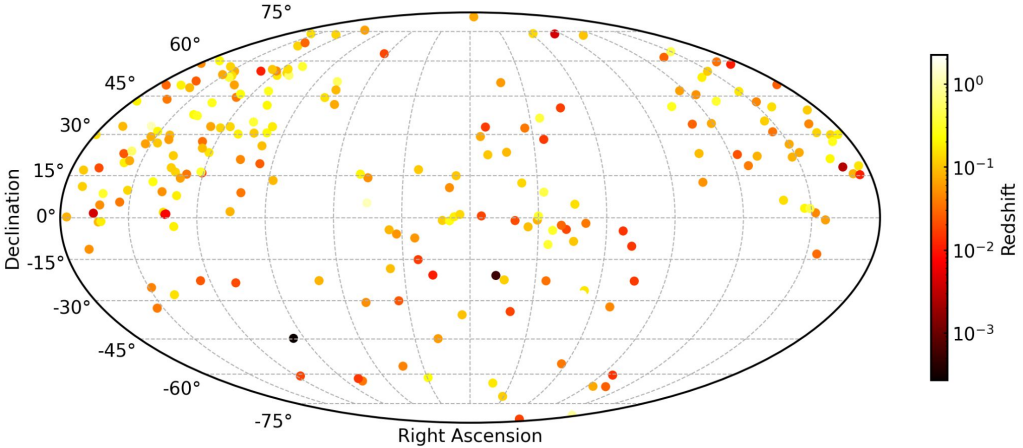
Example slides

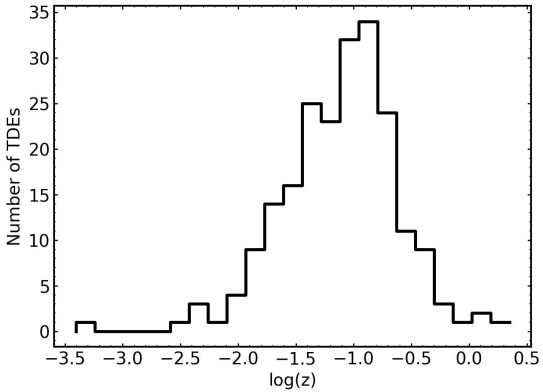
Noah

Dataset Updates

- 218 TDEs right now
- Complete UV, Optical, IR, and Radio dataset
- ~50% of the X-ray data

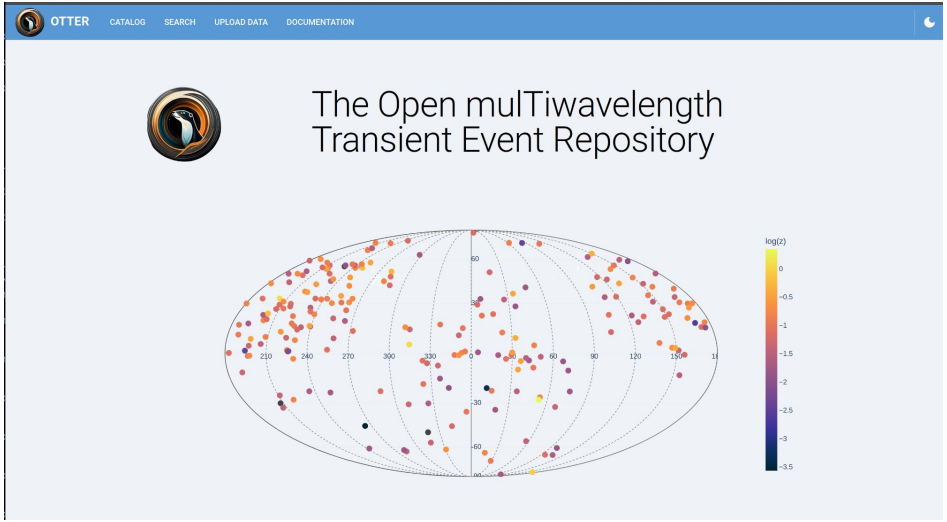


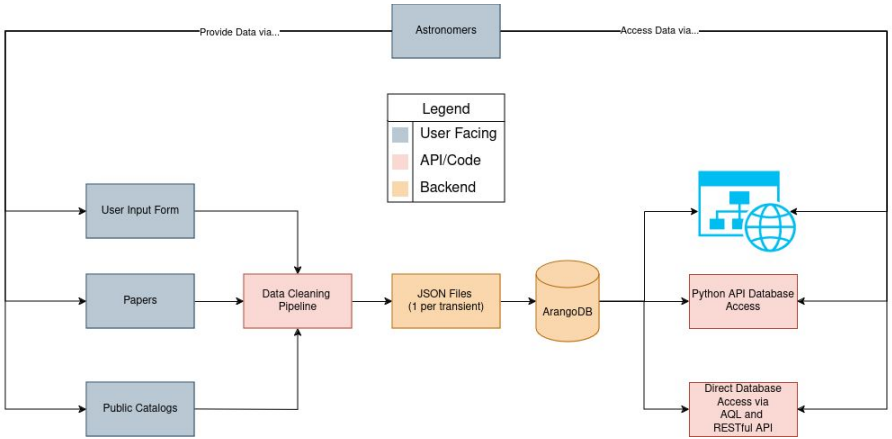




Infrastructure Updates

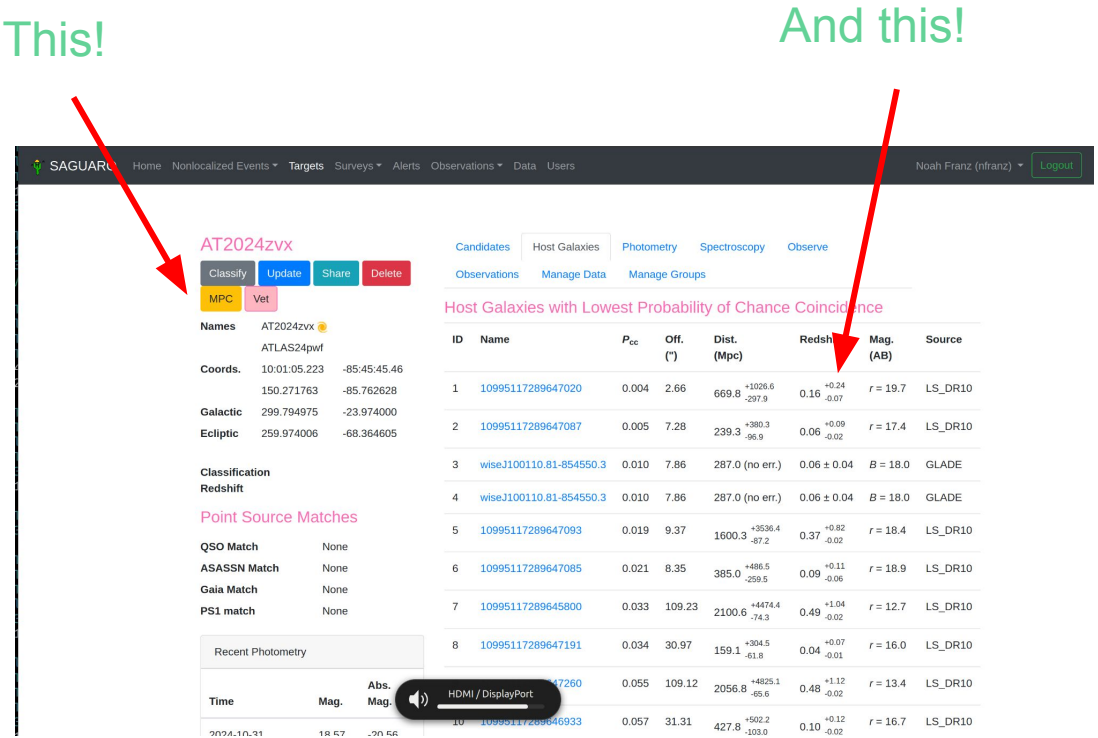
- Backend basically done
- Website nearly done
- API nearly done
- Beginning work with SciServer!





SAGUARO Stuff

- Fixed some minor bugs
  - Deal with asymmetric error bars
  - Fix TNS ingestion
- Implement moving object checker to the vetting code



Future Plans

- Need to solidify the science case for the paper still
  - Either ECLE stuff or modeling stuff
- Write!
  - Two papers or one paper? Opinions?

Goals

- Upload/Publish sometime over the summer
- Oral prelim by end of next september

Example slides

Meredith

Meredith - 01/24/20

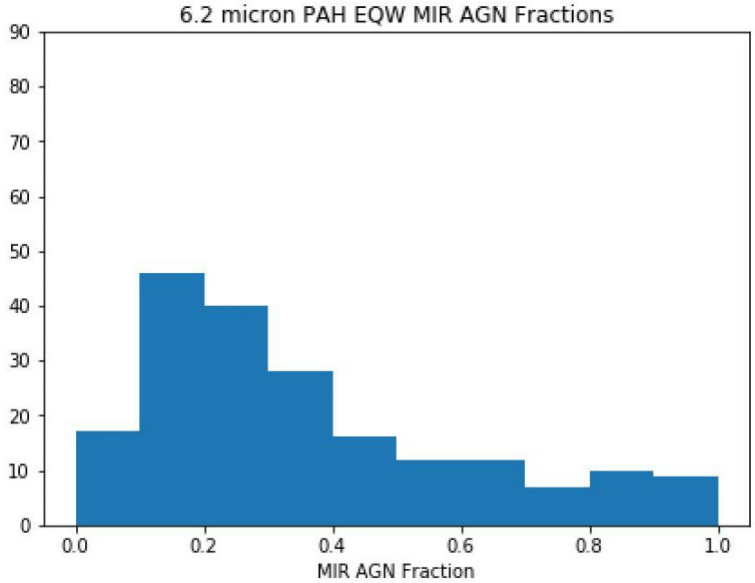
Last Semester: Debugging and verifying Summer 2019 code

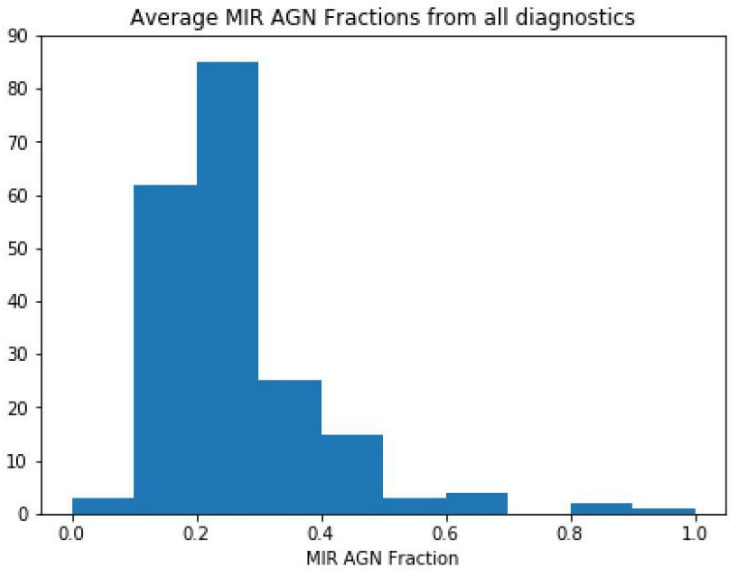
- Bootstrapped errors
- RMS calculation for stack vs. for individual galaxy
- Testing stacking code (+ fitting process) on [SIII] (33.5 microns)

This Semester: Using code to answer science questions

- Comparing [NeII] (12.8 microns), [NeV] (14.3 microns), and [OIV] (25.9 microns) against AGN fraction
- Fewer detections in AGN lines invites stacking

Meredith - 01/30/20

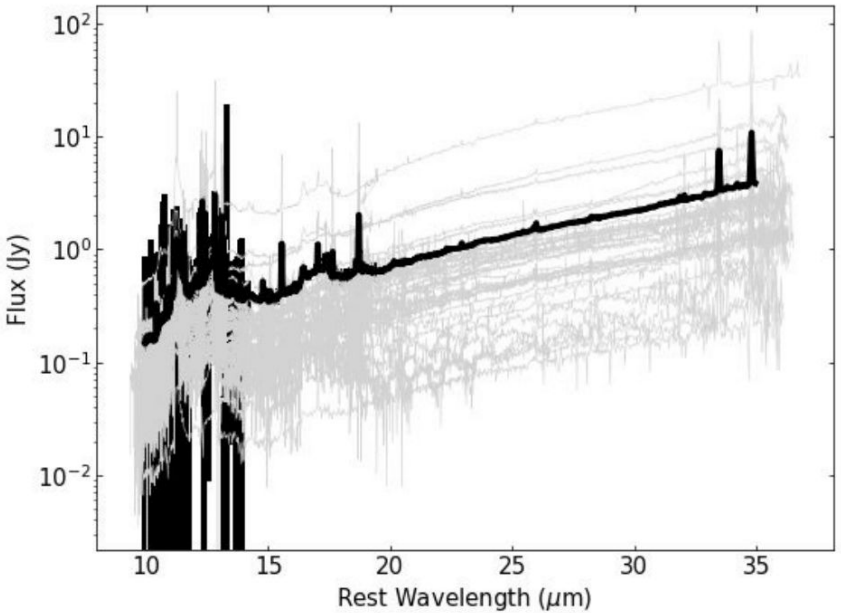


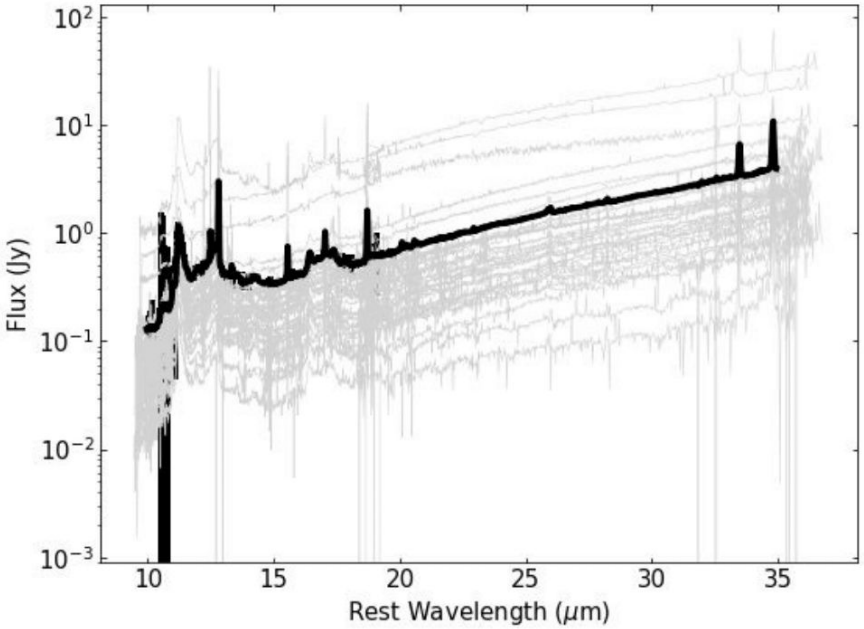


- Input new AGN fractions - very different shapes
- Working on stacking and fitting code
- Next week - determine normalization range, finish fitting code, investigate AGN lines to cut around

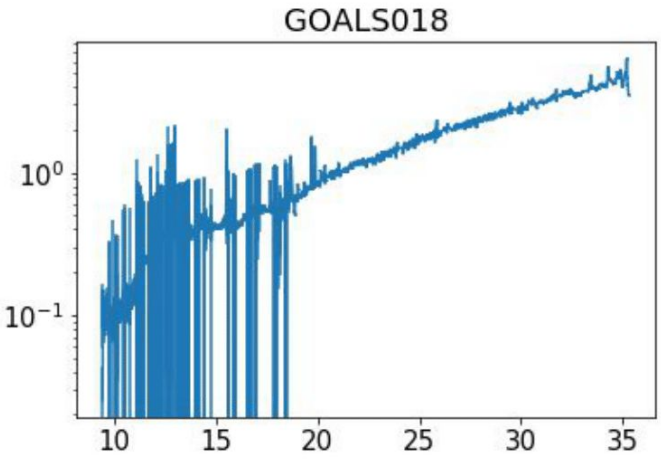
Meredith, 02/07

Bin 1: AGN <.15

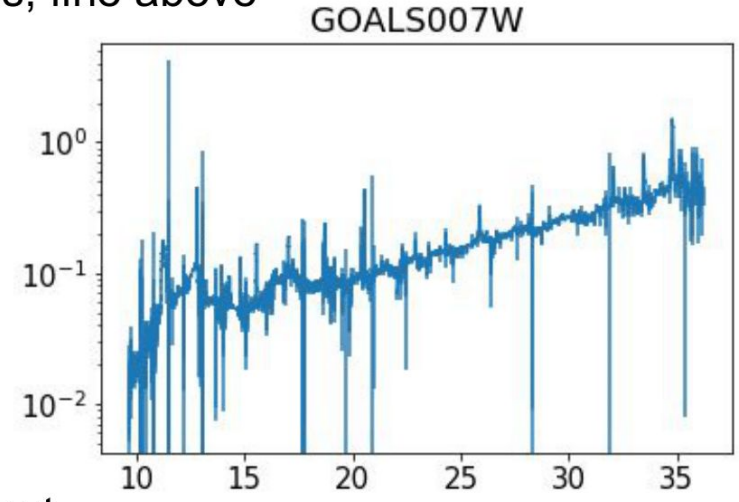


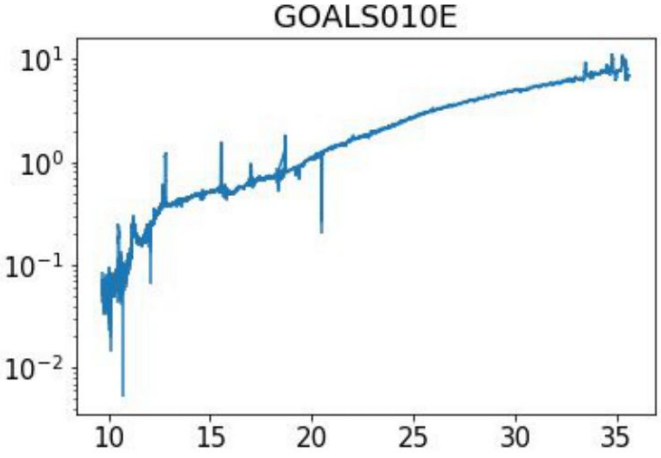


Bin 2: .15 < AGN < .23



Noisy in PAHs, fine above





Fine throughout

Noisy throughout

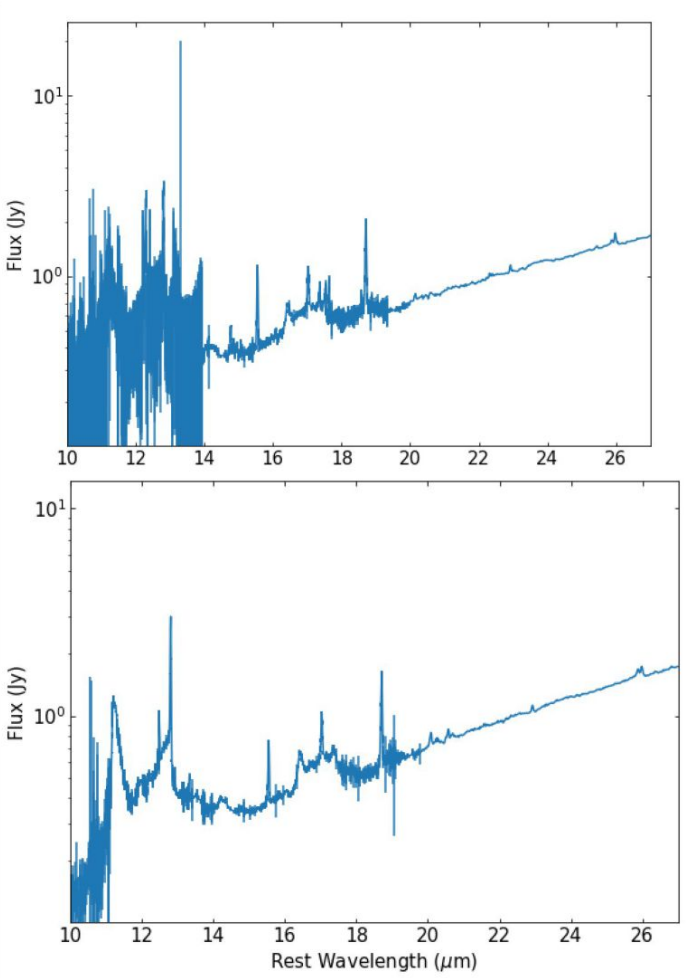
Meredith - 02/14

This week:

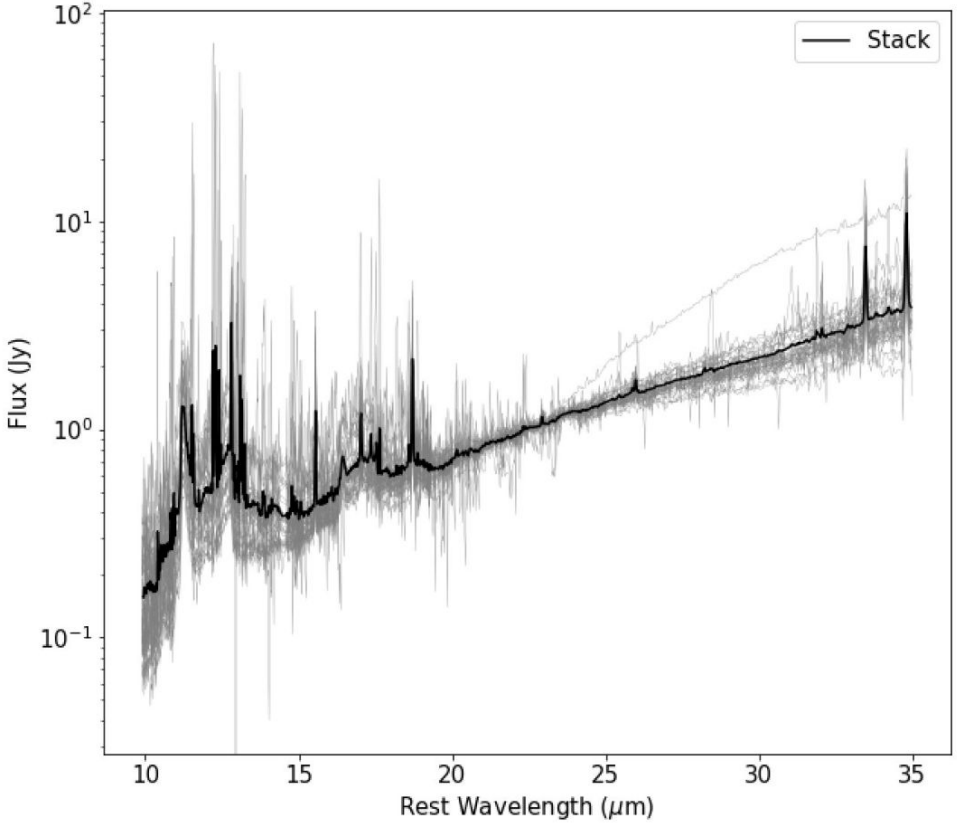
- Checked potential normalization window for lines - nothing
- Added sigma-clipping for the normalization (mean or median?)
- Trying to troubleshoot lowest AGN fraction stack (large errors)
- Finishing REU applications

Next week:

- Make non-continuum subtracted stacks
- Code continuum subtraction
- Start working on fitting

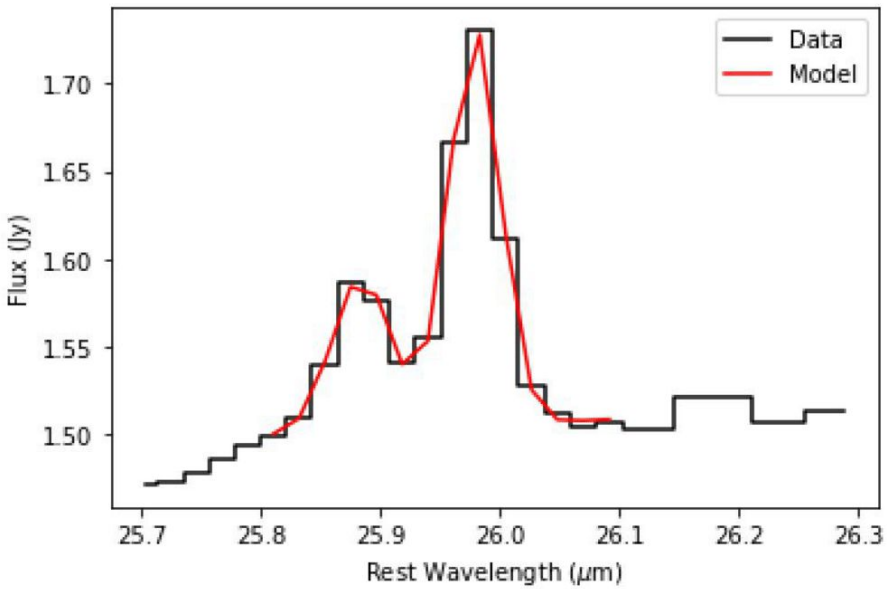


Meredith - 02/28



- Still attempting to find source of Bin 1 noise
- Found and removed obviously wrong galaxy (GOALS066/ IRAS17578-0400) from stack
- Still much noisier than other stacks, need to continue looking at individual galaxies to pick out the noisiest ones (this plot has no error bars)
- Noticed some anomalous galaxies in other bins - need to investigate further

Meredith, 04/10



- Installed/implemented lmfit with help from Jed
- Increased flexibility with the bonus of actually working
- Beautiful results!
- Next week: add luminosity calculation for these two lines (OIV and NeII)
- Much easier than doing it for NeV first, and easier to compare to Inami

What did you appreciate in these updates?

What did you wish they had included?

Discuss with your groups.

How to ask Questions

Where can questions **come from?**

- Questions from reading papers
- Questions from getting stuck
- Questions for next steps

To whom do your questions go?

- Your research professor
- Your research mentor
- Your TIMESTEP grad
- Your peers
- Yourself (& the internet)

Asking **multiple people** often helps, everyone explains concepts differently.

What is one question you currently have about your research project?

Take a minute and write / type your question down.

Anatomy of a question

~~This might be a stupid question but...~~

No question is stupid! If you're asking, it's because you're ready to learn.

Is now a good time to ask you a question?

If you’re asking a question outside of a scheduled research meeting. If the answer is “not now” ask to find a more appropriate time.

I don’t understand this paper.

You might be feeling this way, but this understanding comes slowly. Pick a few things you want to dig into.

I’ve been working through the math in **[paper name],** specifically in **[section]**. Can I review my understanding with you?

Being as specific as you can helps your mentor clue in to where you are mentally.
Asking to talk through papers is normal.

My code isn’t working.

Debugging code with a professor over your shoulder is humbling. Be specific with what help you request.

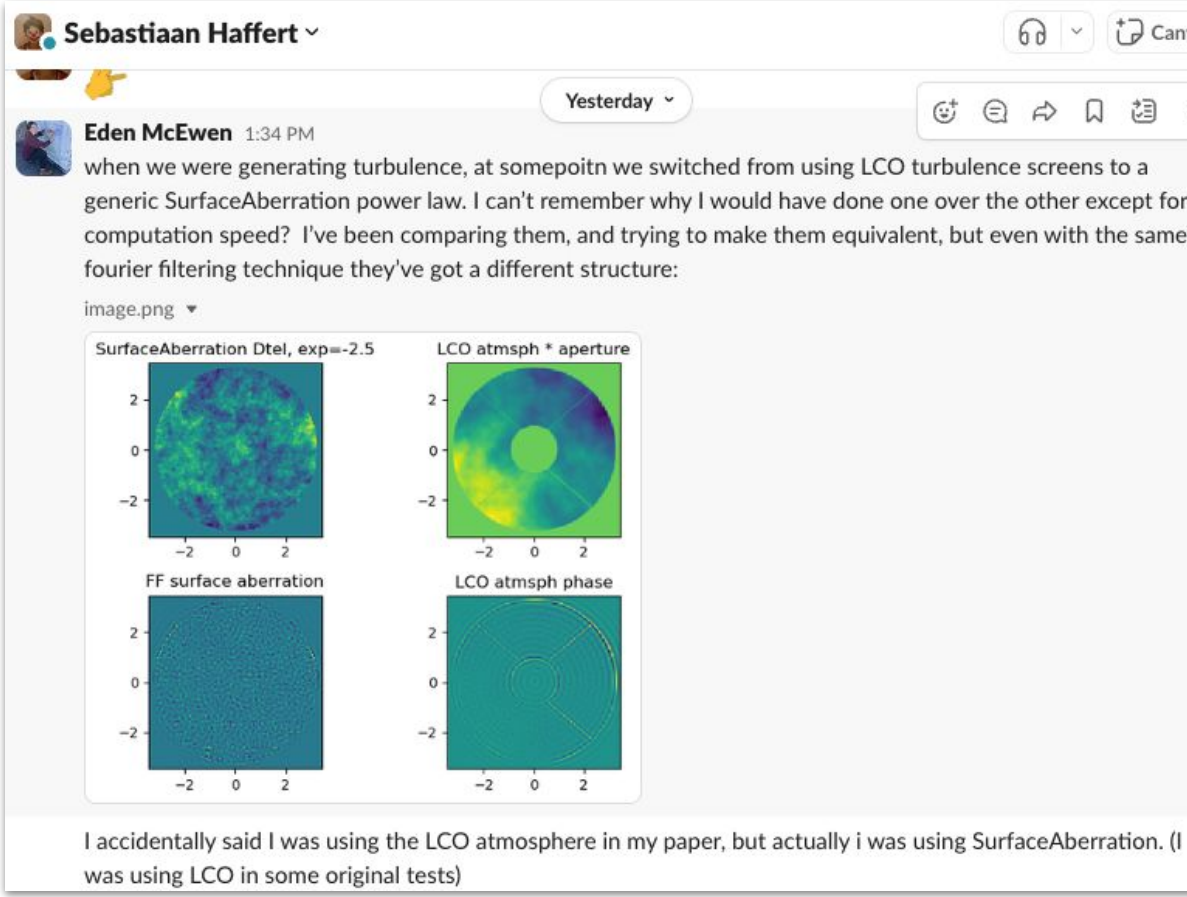
I've been working on code to [**do a thing**], but currently it [**type of error**]. So far I've tried [**debugging steps**], do you have ideas for what I could do next?

Include the steps you've already taken to work through your problem in your question. If you haven't talked in a while, remind your mentor what you've been working on

Wait really?

This might seem excessive, but it’s how I phrase my code questions, especially if over **slack/email**.

I want to make it **as easy as possible** for someone to answer me.



Anatomy of a Question

< Context >

< Description of problem >

< Understanding so far >

< A direct question >

Question Asking - final points

- **Getting stuck** in research is **normal**
  - Your mentors expect and want your questions!
  - It's your mentor's job to help you get unstuck

- A **well phrased question** helps both parties
  - Your mentor understands better the gaps in knowledge and request
  - You're more likely to get a helpful response

- Your **peers** are a resource
  - You're intentionally paired with another student under a mentor (where we could)
  - A TIMESTEP requirement to support each other
  - Explaining to a peer is a great test of your own knowledge

How would you rephrase your question to be more specific?

Take a minute to rewrite your question.

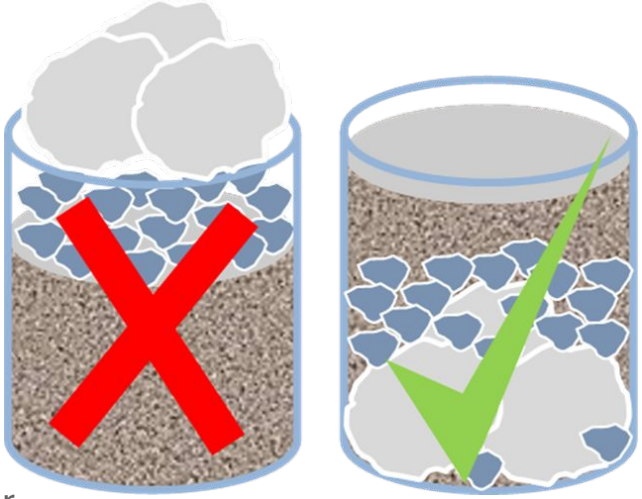
Then pair share with someone in your group the changes you made and why.

Good **notes** = Good **time management**

How notes can improve workflow

- ToDo / Next up lists
  - Creates clear tasks and deadlines
  - Keeps track of smaller tasks
- Clear description of work
  - Gives direction when jumping back in
  - Which makes jumping back in quicker

Pebbles and boulders



Not all tasks are equal.

- **Boulders** - writing paper, math rederivation
  I don't know how to do, far out deadlines

- **Pebbles** - Lab work, another reduction, reading another paper
  I mostly know how to do, near term deadlines

- **Sand** - sending emails, meetings, twiddling plots
  Things that happen regardless, immediate deadlines

Make time where you HAVE to work on your boulders.

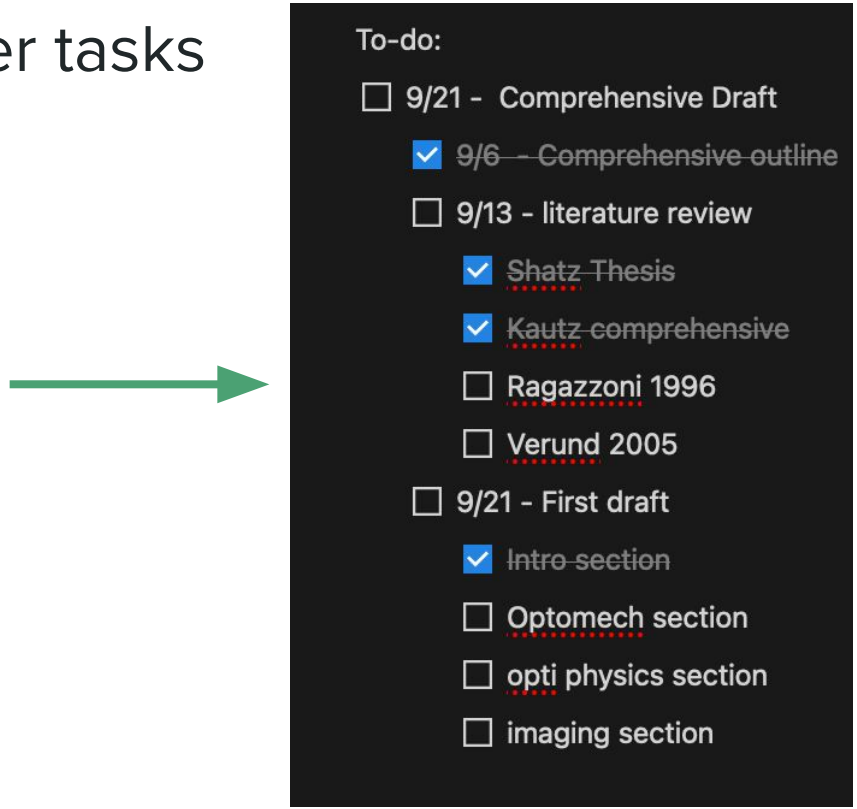
- 30 minutes of dedicated time will break the ice and make the project more manageable.
- Chip away at the larger goal
- They won't fit in your life if you let smaller tasks take up your time

Breaking down larger tasks

To-do:

- [ ] read all references
- [ ] write paper
- [ ] get PhD

I haven't completed any big tasks



But I have completed intermediate ones

Specific goals, achievable tasks (in a day or so), clear deadlines

What other time management skills do you use?

Discuss with your groups.

Meet with your Grad Mentor

Make sure you can access your **research update slides**